# IOT-BASED AGRICULTURAL MONITORING SYSTEM

**BY**

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**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA**

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**JUNE 2025.**

# DECLARATION

We hereby declare that this project titled "**IOT-BASED AGRICULTURAL MONITORING SYSTEM**" was carried out by the following students in partial fulfilment of the requirements for the award of a degree in Bachelor of Engineering at the Department of Electrical and Electronic Engineering, College of Engineering and Technology, Federal University of Petroleum Resources, Effurun.

This work is a record of our original research and has not been submitted previously, in whole or in part, for the award of any degree or diploma in this or any other institution. All sources of information used in this work have been duly acknowledged and referenced.

We accept responsibility for any errors or omissions found in this project.

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## CERTIFICATION

The proposed project work titled **"IOT-BASED AGRICULTURAL MONITORING SYSTEM"** has been duly approved by the supervision of Engr. Mrs. Rebecca Ohwo (Project Supervisor) to be carried out by the students mentioned below and is being submitted for approval to the department of Electrical/Electronic Engineering, College of Engineering and Technology, Federal University of Petroleum Resources, Effurun, Delta State, Nigeria, for the award of the Bachelor's in Engineering in Electrical and Electronics Engineering.

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(HEAD OF DEPARTMENT)

# DEDICATION

This project is dedicated first to Almighty God, whose unfailing grace, wisdom, and strength guided me through every stage of this work. His presence has been my source of direction, perseverance, and success.

I also dedicate this work to my beloved parents and siblings, whose love, prayers, and unwavering support have been the backbone of my academic journey. Your belief in me gave me the courage to push forward even when things became difficult.

Also, to my project Supervisor, Engr. Mrs. Rebecca O. Ohwo.

To all these inspirations, human and digital, I am profoundly grateful.

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# ABSTRACT

This project presents the design and implementation of an Internet of Things (IoT)-based agricultural monitoring system aimed at addressing the inefficiencies of traditional subsistence farming practices in Nigeria. The system integrates an ESP32-WROOM microcontroller with a network of sensors including a DHT11 for atmospheric temperature and humidity, a DS18B20 for soil temperature, and a resistive soil moisture sensor that operates on the principle of inverse electrical resistance, where lower raw ADC values indicate wetter soil and higher values indicate drier soil.

Data is captured through a continuous 2-second sensing loop, with the ESP32 reading all sensors in real-time. The soil moisture sensor is powered only during measurement intervals to minimize electrolysis and prolong sensor life. The captured data drives an automated irrigation control logic based on dual-threshold hysteresis: the pump activates when soil moisture exceeds the DRY_THRESHOLD (2200 raw ADC value, indicating high resistance/dry soil) and deactivates when moisture falls below the WET_THRESHOLD (1600 raw ADC value, indicating low resistance/wet soil). A 20-second safety timeout prevents overwatering in case of sensor failure. All sensor readings and pump states are displayed locally on a 16×2 I2C LCD and transmitted via Wi-Fi every 3 minutes to a Google Apps Script API, which logs the data as timestamped entries in a Google Sheets database.

The system features a custom-built web application accessible from any internet-connected device, providing real-time data visualization, remote manual pump control, and PDF export functionality for farm record-keeping. By combining low-cost hardware, free cloud infrastructure, and intelligent control logic, this project delivers an affordable, scalable, and user-friendly decision support tool that transforms raw environmental data into actionable agricultural intelligence for small-scale farmers.

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# NOMENCLATURE

| Abbreviation | Full Meaning |
|---|---|
| **ADC** | Analogue-to-Digital Converter |
| **AI** | Artificial Intelligence |
| **AO** | Analog Out (Sensor Pin) |
| **API** | Application Programming Interface |
| **B. Eng.** | Bachelor of Engineering |
| **COM** | Common (Relay Pin) |
| **CSS** | Cascading Style Sheets |
| **DC** | Direct Current |
| **DHT11** | Digital Humidity and Temperature (Sensor Model) |
| **DS18B20** | Digital Soil Temperature (Sensor Model) |
| **ESP32** | A low-cost, low-power microcontroller with integrated Wi-Fi and Bluetooth |
| **GND** | Ground (Electrical Reference Point) |
| **GPIO** | General-Purpose Input/Output |
| **HTML** | HyperText Markup Language |
| **HTTP(S)** | HyperText Transfer Protocol (Secure) |
| **I2C** | Inter-Integrated Circuit (Communication Protocol) |
| **IDE** | Integrated Development Environment |
| **IoT** | Internet of Things |

| | |
|---|---|
| **JSON** | JavaScript Object Notation |
| **LCD** | Liquid Crystal Display |
| **LED** | Light Emitting Diode |
| **LM7805** | 5V Voltage Regulator (Linear Monolithic) |
| **NO** | Normally Open (Relay Pin) |
| **SCL** | Serial Clock Line (I2C Pin) |
| **SDA** | Serial Data Line (I2C Pin) |
| **V / VDD** | Volt / Voltage Positive Power Supply |

# CHAPTER ONE

## INTRODUCTION

### 1.1 Background of the Study

The foundation of Nigeria's economy and the main source of income for its citizens is farming. Agriculture has been primarily focused on subsistence for many generations. Large-scale commercial farming has been made possible by technology, but only a small portion of the population can now access it, resulting in a gap between the vast majority of people who engage in small-scale subsistence farming and a small number of commercial farms (FAO, 2025; NBS, 2025).

Although they face many obstacles, this enormous majority, roughly 80% of Nigeria's farmers, are the backbone of the country's agricultural industry (Punch Newspaper, 2025). Their practices are frequently typified by low yields, traditional methods, and scarce resources. The use of manual techniques for crop monitoring and irrigation is one of the main problems with this "old and inefficient" system (NAERLS, 2025).

Due to the frequent overwatering or underwatering of crops, this traditional method directly results in significant water waste as well as high labour waste from manual intervention. Farmers are unable to optimise resource use and are exposed to erratic weather patterns due to their lack of access to real-time monitoring of vital farm parameters like soil moisture, temperature, and humidity (FAO, 2025).

Large-scale mechanisation or credit programs are frequently mentioned as solutions to close Nigeria's agricultural divide, but more urgently, smart, reasonably priced technologies are required. These fundamental inefficiencies can be directly addressed by giving small-scale farmers access to inexpensive monitoring systems, like Internet of Things (IoT) solutions. These systems help modernise subsistence farming from the ground up by providing a way to save water, cut labour costs, and increase crop resilience (International Trade Administration, 2025).

### 1.2 Statement of the problem

In Nigeria, manual crop monitoring techniques are crucial to subsistence farming. Because crops are frequently either overwatered or underwatered, this conventional method is ineffective and directly results in large water waste, which has a substantial impact on crop yields (IFAD, 2023). According to studies, a significant amount of agricultural water is wasted because of inadequate monitoring and erratic watering schedules, particularly for smallholder farmers (Trade.gov, 2025).

According to Amnesty International (2025), manual crop monitoring also increases labour waste, lowers farmer returns, and increases the likelihood of food shortages, especially in areas where insecurity already interferes with farming operations and rural livelihoods. The impact of inefficiencies among small-scale farmers has gotten worse, according to recent national reports, as the cost of food production and general agricultural inputs has increased significantly (Guardian.ng, 2025; NAERLS, 2025; Punch Newspaper, 2025).

Lack of access to reasonably priced, real-time monitoring of important farm parameters is the main issue. Farmers cannot optimise water use for irrigation or react appropriately to changing weather conditions without consistent and reliable data. Local food production and long-term agricultural sustainability are greatly impacted by the disconnect between the need for effective farm management and the basic, manual tools that the majority of farmers have access to (IFAD, 2023).

This study suggests creating and deploying an Internet of Things-based agricultural monitoring system that provides real-time environmental and soil moisture data in order to solve this issue. An automated watering mechanism intended to maximise resource utilisation, minimise waste, and enhance crop resilience will be made possible by this system.

### 1.3 Aim and Specific Objective

The aim of this project is to design and implement an IoT based Agricultural Monitoring System. The specific objectives are:

(i). To develop a user-friendly web application that serves as an interface for the remote monitoring and control of the farming system.

(ii).To implement an automated irrigation system that activates a water pump based on real-time data collected from a soil moisture sensor, ensuring optimal water usage.

(iii). To integrate a network of sensors for the continuous, real-time monitoring of key environmental parameters, specifically soil moisture content, temperature, and humidity.

(iv). To enable real-time data visualization within the web application, presenting sensor data through intuitive graphs and charts to help track environmental conditions over time.

### 1.4 Scope of Study

The work is limited to the design and construction of an IoT-based agricultural monitoring system for smart farms, primarily intended for small-scale and subsistence farming, is the main focus of the study. This strategy is in line with ongoing studies that support affordable IoT solutions that are appropriate for Nigerian smallholders (Udom & Ekanem, 2024). Targeting small-scale farmers who use manual crop monitoring techniques, the system is designed especially for use in Delta State.

The technical scope focuses on integrating an ESP32-WROOM microcontroller with a network of sensors, such as a DS18B20 soil temperature sensor, a DHT11 air temperature and humidity sensor, and a resistive soil moisture sensor. Recent studies have successfully implemented similar ESP32-based agricultural systems, proving their suitability for low-cost environmental monitoring (Abdulrahman & Alaka, 2023; Okomba et al., 2023; Rajput, 2025).

The software component entails creating a unique solution to cut costs by eschewing premium IoT platforms like ThingSpeak. Instead, a Google Apps Script API is used to transfer sensor data from the ESP32 to a Google Sheet, which serves as a free cloud-based database. Related studies that use Google Sheets as a low-power IoT backend for agricultural applications have validated this approach (Chiagunye, 2025). This data is retrieved and displayed for the user by a custom web application that was created with HTML, CSS, and JavaScript.

## 1.5 Scope and Limitations of the Study

### Scope of the Study

The scope of this study specifically includes:

(i). The design and assembly of the IoT hardware unit.

(ii).The development of a web interface for remote monitoring.

(iii). The implementation of an automated watering system based on soil moisture data.

(iv). Real-time data visualisation of sensor readings (temperature, humidity, and light) on the application.

### Limitations of the Study

This study is limited to the aspects mentioned above and does not cover:

(i). Advanced data analytics, such as AI-based crop yield prediction or disease detection.

(ii). The automation of other farming activities, including pest control or fertilization.

(iii). The manufacturing of electronic sensors.

(iv). Large-scale field deployment or the commercial production of the system.

(v). No user authentication

(vi). In-depth cybersecurity protocols beyond basic user authentication.

(vii). The creation of an alert system to notify users of predefined environmental conditions.

## 1.5 Significance of the Study

This study is significant because it takes a direct approach to addressing the pressing issues of water waste and ineffective farm management in conventional agriculture. For individual farmers, the environment, the larger agricultural industry, and upcoming researchers, this project offers an affordable, workable solution with numerous advantages.

The following are the particular advantages:

**Advantages for Farmers**

(i)**. Lowers Labour and Costs:**

The automated irrigation system saves the farmer a great deal of time and money by doing away with the need for manual watering.

(ii). **Enhances Crop Yield:**

The system produces a stable environment for crops by maintaining ideal soil moisture and avoiding both over- and underwatering, which can result in higher yields and farm returns.

(iii)**. Enables Remote Management:**

Farmers can monitor and manage their crops from any location with the help of the web-based application, which enables them to respond to issues immediately without having to be there in person.

**Environmental and Agricultural Sector Benefits**

(i). **Water Conservation:**

The main purpose of the system is to maximise water use, which significantly lowers water waste and encourages environmental preservation.

(ii). **Modernisation of Small-Scale Farming:**

By offering a reasonably priced and easily accessible gateway to "smart farming" technology, this project contributes to the modernisation of small-scale farming methods in Nigeria.

(iii). **Climate Change Resilience:**

The system assists farmers in anticipating and adjusting to the erratic weather patterns linked to climate change by giving them up-to-date information on the state of their farms.

# CHAPTER TWO

# LITERATURE REVIEW

## 2.1 Overview of IoT-based agriculture monitoring system.

An IoT-based agricultural system uses a network of various sensors, field devices, and indoor devices, with software for the collection, transmission, and analysis of real-time data from the field. With this data, farmers can make decisions and increase the efficiency of their produce.

The primary goal here is to move farming practices from a more reactive practice (a system in which problems are dealt with after they occur) to a more proactive one (dealing with and anticipating problems before they occur). Farmers will know their crops, soil, and environmental conditions, which will, in turn, boost crop yields, lower labour costs, increase decision-making, and reduce risk.

Key areas in an IoT-based agricultural monitoring system are the sensors, network connectivity devices for transmitting data from sensors to a cloud-based system. There's also some form of data processing. In some systems, machine learning algorithms are used before relaying decisions back to actuators to make effective decisions, and also to the end users in the form of a graphical user interface (GUI), which the user has some form of control depending on the way the system was implemented; it can be fully automated or just semi-automated. There is also the provision of some form of data visualisation, interpreting data into graphs and charts for easy understanding by the end users.

## 2.2. Review of Related Literature

Over the past few years, several researchers have explored IoT applications in agriculture. Their studies range from soil moisture tracking to crop health monitoring. Despite this, there exists a gap in developing scalable and low-power intensive systems.

This literature review will focus on IoT-based agricultural monitoring systems, examining how they enhance productivity as well as their drawbacks and limitations. Some notable studies are highlighted below.

**Hashim et al. (2015)** conducted a study that reviewed various approaches for making a low-cost, flexible agriculture monitoring system. He proposed using an Arduino UNO with an Ethernet shield with the capability to function as an embedded micro-web server, which can be accessed via IP. The primary aim was to monitor temperature and soil moisture, with control and data visualisation through a mobile app. The inclusion of an embedded web server suggested a potential for web accessibility. It had a dedicated user interface which could be accessed via their Android app. The system relied on a specific mobile platform rather than a universal web-based solution. This highlights the continued preference in some research for mobile apps over fully integrated web platforms, leaving a gap for universally accessible, browser-based monitoring and control systems.

**Liu Dan et al. (2015)** developed a system focused on agricultural greenhouses, which differs significantly from systems designed for subsistence farmers, the primary focus of our work. Their design utilised ZigBee wireless technology, employing CC2530 chips for sensor nodes used in data collection. Data transfer was conducted wirelessly, but the system's only connection to external devices was through a direct serial port linked to a PC. The system operated as a local closed-loop system, meaning data could not be accessed remotely via the Internet. Consequently, farmers or staff had to be physically present with the connected computer to access and analyse farm data. The system's dependence on an on-site computer and lack of internet connectivity made it unsuitable for subsistence farmers, who require a low-cost, remotely accessible monitoring system.

**Mohanraj et al. (2016)** proposed a system that went beyond normal IoT-based systems by introducing the use of a Knowledge Base (KB). They prototyped their system on a TI CC3200 Wi-Fi microcontroller, integrating real-time sensor data with the KB of crop information and cross-referencing it with market prices to provide farmers with insights for making profitable decisions. While this was a smart and innovative improvement over typical IoT-based agricultural monitoring systems, the system faced challenges due to its high cost and complexity, which made implementation difficult.

**Gondchawar and Kawitkar (2016)** developed an ambitious smart farming system made up of three main parts: a GPS-controlled robot for weeding and spraying, a smart warehouse for managing crop storage, and a smart irrigation unit. The setup used a Raspberry Pi as the main

server to connect all parts and enable internet access. The authors explained that farmers could control the system using an Android app or computer commands. However, this design is expensive and complicated. With features like a mobile robot, smart warehouse, and multiple Raspberry Pi units, it is not suitable for small-scale or subsistence farmers. The study reflects a common issue in research: creating systems packed with features but too complex or costly for the people who need them most.

**Suma et al. (2017)** proposed integrating an IoT-based farming system in response to the increasing migration from rural to urban areas, which results in fewer people available for subsistence farming. He identified the impending shortage of manual labour as more individuals move to urban centres. In his research, he employed a PIC16F877A microcontroller to process data from various sources, including soil moisture and temperature sensors, with the addition of a PIR sensor for intruder detection. The system functioned primarily as a prototype, capable of collecting farm data and transmitting it via GSM alerts to a mobile application. This approach provided a low-cost and integrated solution for environmental monitoring and crop protection. However, despite its advancements, the study exhibited limitations in several areas — particularly its heavy reliance on a mobile application, which restricts accessibility compared to a universal web-based platform that can be accessed from any operating system, including Android, Windows, Linux, or macOS. Adopting a unified, cross-platform system would effectively reduce the overall cost of developing separate applications for different ecosystems.

**Akkaş and Sokullu (2017)** designed a wireless sensor network for greenhouse monitoring using special MicaZ motes that communicate through a ZigBee-based protocol. These motes, which must be programmed using TinyOS and nesC, collect data on temperature, light, pressure, and humidity. The information is sent wirelessly to a MIB520 base station, which connects via USB to a computer. Although the authors mentioned that farmers could access the system through mobile phones or computers online, the actual prototype depended on local software called MoteView installed on the connected PC to view and store the data. This system has two main drawbacks for small-scale farmers. First, the specialized MicaZ motes are more expensive and complicated compared to simpler devices like Arduino or ESP boards. Second, because it depends on a dedicated PC running specific software, it lacks the flexibility and easy remote access offered by cloud-based web applications.

**Rajakumar et al. (2018)** focused on designing an IoT-based agricultural monitoring system to address the issue of water wastage in India and to provide a low-cost automated solution. For his system design, he used an Arduino Uno, a relatively cheap microcontroller, interfaced with various sensors such as a temperature sensor, humidity sensor, and ultrasonic sensor. A key element of his system was the integration of an ESP8266 Wi-Fi module, which has the ability to transmit all sensor data to a prebuilt application, ThingSpeak. His approach broke the limits of using basic, traditional, prebuilt applications by introducing a more web-based platform where farmers could remotely monitor farm parameters and visualize them in graphs or charts, an advantage for those who lack basic reading skills. However, ThingSpeak limits functionality mainly to data visualization and is not very customizable to user preferences. The integration of more complex functions or the addition of user-specific features would be practically impossible.

**Putjaika et al. (2018)** used LoRaWAN (Long Range Wide Area Network) in their design, focusing primarily on large remote areas or areas with low network connectivity. They used a sensor node, built with an Arduino and a LoRa module, to collect data on soil moisture and air temperature. The gathered data is then transmitted over a long distance to a central LoRaWAN gateway, which uploads this data to the Netpie IoT cloud platform. The system utilises a mobile application for visualisation and for controlling a water pump. The system has the advantage of long-range capability, which solves the connectivity issues faced by Wi-Fi or Bluetooth-based systems. However, implementing a system with LoRaWAN is significantly more expensive and complex than standard Wi-Fi or GSM systems. This makes the system too costly for many subsistence farmers. Furthermore, the complexity of installation requires people with significant technical know-how. Another critical factor is the system's core dependence on the Netpie cloud platform and a dedicated mobile application. These services likely require an active data subscription, further increasing the overall cost and making the system unaffordable for most subsistence farmers.

**Rajinder Kumar M. Math and Nagaraj V. Dharwadkar (2018)** proposed a low-cost IoT-based weather station designed to support precision agriculture in India. They recognised that unpredictable weather patterns contribute significantly to crop losses and observed that many Indian farmers, including those in rural areas, now have access to high-speed internet. Their system aimed to deliver localised, real-time weather data to assist farmers in making informed

decisions about irrigation and harvesting. However, the abstract of their study primarily discusses the importance of real-time data and the design of the data collection system, while neglecting to describe the user interface. It remains unclear how farmers were intended to access this information; whether via a web portal, mobile application, or SMS notifications. This omission reveals a critical gap: a low-cost data acquisition system can only achieve its purpose if complemented by an equally affordable and accessible monitoring platform, such as a cloud-based web application.

**Khattab et al. (2019)** developed a specialised IoT system with the primary focus on detecting early plant disease with the help of Artificial Intelligence (AI). Their hardware consists of a custom-built, solar-powered agro-weather station equipped with various sensors (air/soil temperature and humidity, leaf wetness, rain, wind, solar radiation). It transmitted data via SMS. The main component of their system was an AI expert system capable of processing sensor data based on pre-developed disease models (e.g., for tomato/potato blight) to predict outbreaks. Users were able to access the system's outputs and warnings through a web-based Graphical User Interface (GUI). The system incorporated a web interface and claimed lower hardware costs compared to commercial alternatives (cost optimisation was not a primary goal). Its complexity and reliance on custom hardware, as well as on sophisticated AI components, and its main focus on disease prediction, which in reality was not the main focus of subsistence agricultural farming, and its accessibility for less technical people made it limited mainly to subsistence farmers.

**Pathak et al. (2019)** developed an IoT-based farming system using an Arduino platform to measure temperature, turbidity, pH, and soil moisture. The sensor data was displayed through the ThingSpeak cloud platform. Instead of focusing on real-time monitoring, the system used this data as input for a Cuckoo Search Algorithm to predict and recommend suitable crops for the soil. While this idea is innovative, it has two main drawbacks for small-scale farmers. First, it depends on the basic, non-customizable ThingSpeak interface for viewing data. Second, it emphasises complex prediction algorithms rather than offering a simple and practical monitoring tool that farmers can easily use to make quick decisions.

**Rajesh et al. (2020)** proposed a low-cost hybrid monitoring system that implements a GSM module for sending critical SMS alerts directly to farmers' phones, and a Wi-Fi module for

transmitting sensor data—such as temperature and humidity—to an existing platform like ThingSpeak for data visualization. While this hybrid approach allows farmers to access both a web platform like ThingSpeak and receive quick SMS alerts, it disrupts the user experience, as the farmer has to switch between the ThingSpeak platform and regular SMS messages to obtain farm data. Although this system appears to address most user requirements, it has a drawback in terms of flexibility. A system specifically designed according to the user's needs would be more reliable than relying on a prebuilt model, which might also require a monthly subscription.

**Nandhini et al. (2020)** developed an IoT-based system for making agricultural practice more precise. He focused on monitoring soil nutrients (Nitrogen, Phosphorus, Potassium - NPK), pH, temperature, humidity, and soil moisture. His system used a Raspberry Pi 3 Model B as the central processing unit. The collected data from the sensors is transmitted to a cloud server, and farmers were able to access this information through a web page plus a mobile application. His system also had the feature of automated irrigation which activates a water pump when the soil moisture level drops below a predefined threshold. The system provided a comprehensive monitoring solution plus a user-friendly web and mobile interface. It used a Raspberry Pi with other specialized sensors (like NPK and pH sensors), which made his system relatively very expensive. Subsistence farmers could not possibly afford it. Furthermore, the maintenance of a Raspberry Pi and a cloud server backend requires a higher level of technical expertise, which most subsistence farmers lack. The system depended on a stable internet connection for real-time data transmission to the cloud.

**Aman Jain and Abhay Kumar (2020)** proposed a smart agriculture system utilizing an ARM7 processor to monitor key environmental parameters such as temperature, humidity, soil moisture, and water levels. Their paper highlights the advantages of IoT in agriculture, referencing elements such as web-based surveys, remote monitoring, and an Android application. However, the study remains notably vague regarding the implementation of the user interface, as the presented results only display data on what appears to be a local PC monitor. This ambiguity between the intended use of a web platform, mobile application, or local interface, combined with the reliance on a complex ARM7 processor, suggests a lack of a clear, low-cost, and accessible solution for farmers. Consequently, the work fails to establish a coherent or user-

friendly data access method, underscoring the need for a well-defined, simplified web-based platform for effective agricultural monitoring.

**Ahmad Faisol Suhaimi et al. (2021)** developed a system primarily focused on creating a low-cost solution for automation and security. The study addressed issues related to ineffective watering and intrusion. The researchers used an ESP32 microcontroller for data collection, which included parameters such as humidity, temperature, and soil moisture content. They also implemented an automation layer using a Real-Time Clock (RTC) for irrigation scheduling and an intrusion detection system. The security system consisted of a PIR sensor connected to an ESP32 camera with a buzzer, enabling the system to capture images of intruders. The overall design of the system was low-cost and featured essential functionalities; however, it had several limitations. It relied heavily on an Android application, required manual monitoring of data and pump control, and the buzzer could only be activated through the Android app. This platform-specific approach limited accessibility, as users on iOS or Windows devices could not access the system.

**Nagasubramanian et al. (2021)** developed an advanced crop disease detection system that uses IoT sensors and cameras to gather data, which is then analysed with machine learning (ML) methods such as ensemble nonlinear SVM (ENSVM) and CNN. The system aims to recognise patterns on plant leaves to detect diseases early. However, like the studies by Siddiquee et al. (2022) and Khattab et al. (2019), this approach is both expensive and complex. It focuses on high-level ML processing that requires powerful computing resources and technical skills, making it impractical for small-scale farmers who mainly need simple tools to monitor their crops and environment. In addition, the paper provides little detail about a user-friendly interface, mentioning only “terminal IoT components” that send notifications, which limits its accessibility for everyday farm use.

**Gaikwad et al. (2021)** developed a low-cost precision farming system that includes an Arduino-based IoT device, a smartphone app, and a web server. The system tracks soil moisture and temperature, as well as air humidity and temperature. A key strength of their design is that the smartphone app serves as both the user interface and a data collection tool, using the phone’s GPS to record field locations—eliminating the need for a separate GPS module. However, this setup has an important drawback: it depends entirely on an Android app. This makes the system

limited to Android users, leaving out farmers who use other devices. It also creates a split user experience, where farmers must use the app for fieldwork and the web platform for viewing results. This shows the need for a single, accessible web-based system that can handle both data collection and analysis on any device.

**Siddiquee et al. (2022)** focused more on developing advanced machine learning algorithms for IoT monitoring systems. He used a Convolutional Neural Network (CNN), a machine learning algorithm designed for processing images, videos, and similar data. He was able to use these data to qualify the viability of vegetables. His main focus was on tomatoes specifically. What he basically did was, using these data of different tomato crops, he was able to build a machine learning algorithm capable of telling farmers when a single tomato fruit was ripe and also detect various defects in the crop. He used a robot mounted with a camera module to get this data. His research proved that using the CNN method in assessing the viability of crops outperformed other traditional methods. His predictions were able to reach a staggering efficiency of 90%. However, his system was nowhere near low-cost or budget-friendly. Building and implementing machine learning algorithms on a large scale, as he intended to do, requires a great amount of capital and a large sample of data. Farmers who could afford this kind of system would be more on the commercial side. Also, his system is not something anyone without a background in machine learning or artificial intelligence would be able to use or implement in their farming practices. Not only would a farmer be paying for the system, but they would also have to make provisions for hiring workers in that field or training their own staff.

**Paschal Uchenna Chinedu et al. (2022)** designed an IoT-based monitoring system using an Arduino Uno in its collection of data on various environmental parameters like temperature, humidity, soil moisture, and soil pH. Communication was based on a GSM module, capable of transmitting sensor readings to the ThingSpeak IoT platform for basic remote visualisation and analysis. The system is also designed to send SMS or email alerts to the farmer when sensor values, particularly soil moisture, cross predefined thresholds. This approach aligns with the low-cost objective by using Arduino and standard sensors. However, similar to Rajesh et al. (2020), it necessitates a fragmented user interaction model: alerts are received via SMS/email, while data trends must be viewed separately on the ThingSpeak website. The sole reliance on GSM for data

transmission might also introduce ongoing operational costs, contrasting with systems utilising potentially free local Wi-Fi.

**Ajit Divakaran Mini et al. (2023)** suggested a cost-effective IoT-based robotic system with the sole aim of optimising resource use, such as fertiliser and water. In his design, he uses an ESP32 for sensing soil and environmental parameters (temperature, humidity, soil moisture) and Wi-Fi communication. It is able to send data to the ThingSpeak IoT platform, which has the feature for storage and analysis. He used a separate Arduino Nano to handle the robot's movement (DC motors). His system featured both control and monitoring interfaces: historical data can be visualised using the ThingSpeak platform, while sensor readings and robot control were handled via a mobile application connected through a Bluetooth module (HC-05 module). This approach maintained low hardware costs and also incorporated a web-based platform for data logging. It gave users the inconvenience of switching between two platforms for proper data gathering. The reliance on Bluetooth significantly limited the system’s range of operation; controlling the robot and viewing real-time data would require both the system and the user to be in very close range, unlike a fully web-integrated system with a wider range.

**Indira et al. (2023)** developed an agricultural monitoring system that combines IoT and Artificial Intelligence (AI) technologies. Their design focused on collecting environmental data, such as temperature, humidity, and soil moisture, using IoT sensors, which were then processed with AI techniques to analyse and improve farming decisions. The study demonstrated how smart technologies can support modern agriculture, but it also has some challenges for small-scale farmers. The system’s AI-based setup requires advanced hardware, reliable internet access, and technical knowledge, making it costly and difficult to use in low-resource environments. While it shows the potential of combining IoT and AI in farming, it is better suited for research or large-scale agricultural operations than for everyday use by subsistence farmers who need simpler and more affordable monitoring tools.

Our review of these 21 sources indicates a strong trend toward using IoT for agricultural monitoring. However, the existing solutions diverge into several categories, all of which leave a significant gap for a platform that is simultaneously low-cost, user-friendly, and provides a unified, cloud-based web interface. Many systems rely on fragmented interfaces: SMS/Email alerts (Suma et al., Khattab et al., Rajesh et al., Paschal Uchenna Chinedu et al.), dedicated

mobile apps with limited accessibility or range (Suma et al., Ajit Divakaran Mini et al., Hashim et al., Putjaika et al., Suhaimi et al., Gondchawar & Kawitkar, Indira et al., Gaikwad et al.), basic web graphing platforms like ThingSpeak (Rajakumar et al., Rajesh et al., Ajit Divakaran Mini et al., Paschal Uchenna Chinedu et al., Pathak et al.), or systems tethered to a local PC (Liu Dan et al., Akkaş & Sokullu, Jain & Kumar). This forces users to juggle multiple platforms. Furthermore, many proposed systems fail the "low-cost" and "low-complexity" requirements for subsistence farmers. High-end systems use costly components like machine learning and AI (Siddiquee et al., Khattab et al., Indira et al., Nagasubramanian et al., Pathak et al.), advanced sensors and Raspberry Pi (Nandhini et al., Indira et al., Gondchawar et al.), specialised communication technologies like LoRaWAN (Putjaika et al., Indira et al.), complex advisory backends (Mohanraj et al.), or multi-part robotic systems (Gondchawar & Kawitkar, Ajit Divakaran Mini et al.). Even systems with custom web interfaces (Khattab et al.) are either part of highly specialised AI platforms or are limited to local network access, failing to provide true remote monitoring. The work of Math & Dharwadkar (2018) and Jain & Kumar (2020) further confirms this gap by focusing so heavily on data collection that the method of data delivery to the farmer is either unspecified or poorly defined. There remains a clear opportunity to develop a solution that integrates the most affordable hardware (like ESP microcontrollers) with a custom-built, **cloud-hosted** web application designed to handle all key monitoring aspects within a single, universally accessible interface.

The review of 21 scholarly works (2015-2023) reveals that existing IoT agricultural systems fall into four categories with critical limitations: fragmented interfaces forcing farmers to juggle multiple platforms (Rajesh et al., 2020; Chinedu et al., 2022), platform-dependent mobile applications excluding non-Android users (Hashim et al., 2015; Suhaimi et al., 2021; Gaikwad et al., 2021), high-cost systems incorporating expensive components like Raspberry Pi or AI/ML algorithms unaffordable for subsistence farmers (Nandhini et al., 2020; Siddiquee et al., 2022), and locally-tethered setups requiring on-site computers for data access (Liu Dan et al., 2015; Akkaş & Sokullu, 2017).

This project directly addresses these identified gaps through key innovations: a unified web interface accessible from any internet-connected device eliminates platform fragmentation and operating system dependency; low-cost hardware selection using ESP32 microcontroller and

basic sensors with free Google Sheets cloud storage makes the system affordable at ₦145,215; true remote monitoring via Wi-Fi transmission enables anywhere-anytime access; and value-added features including manual pump override, twenty-second safety timeout, and PDF export for farm records provide practical utility absent in existing systems. By combining affordable hardware with a custom-built, universally accessible web application, this project delivers an accessible, affordable, and unified decision support tool tailored for small-scale Nigerian farmers, directly filling the gap identified in the literature for a system that is simultaneously low-cost, user-friendly, and remotely accessible from any device.

### 2.3 Materials Used

This section provides a detailed breakdown of the primary components selected for this project, along with the rationale for their use.

#### 2.3.1 Prototyping and Circuit Assembly

(i). **Breadboard:** This is an 8-point solderless board used in the initial prototyping of the system circuit. This allowed us to test, debug and make changes in our circuit without permanently soldering physical components.

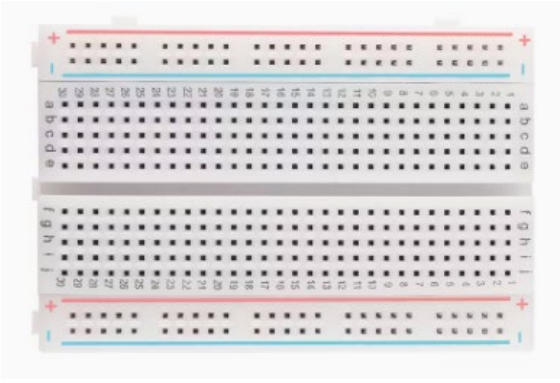


Figure 2.1 Breadboard

(ii).**Jumper Wires:** Male-to-male and male-to-female jumper wires, these were used to make temporary connections to the controller and various components.



Figure 2.2: Jumper Wires

(iii). **Veroboard (Stripboard):** Like the breadboard, but used to make permanent connections. Soldering led and iron are used to make this connection. Where each individual component is solder permanently to the board.

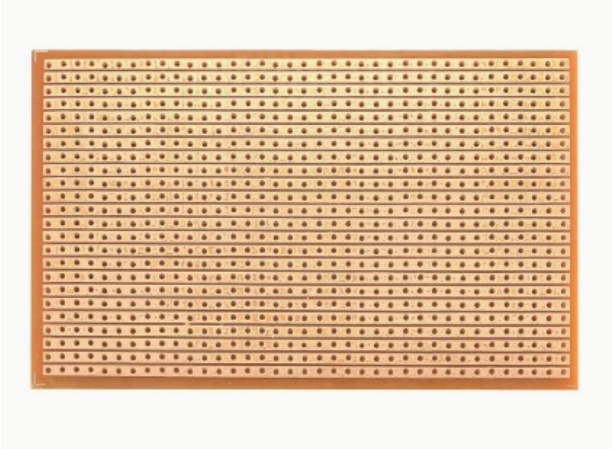


Figure 2.3 Veroboard

(iv). **Soldering Iron and Solder:** The soldering iron is a 60watt device use to apply solder to our connection to make this connection permanent.



Figure 2.4 Soldering Iron




Figure 2.5 Soldering Lead

(v). **LED and Resistor:** A standard 5mm LED was included as a "power on" indicator. A 220-ohm resistor was used in series with the LED to limit the current and prevent it from burning out.



2.6 Figure LED

(vi). **Buzzer:** This is a 5v system that sends auditory notification to nearby farmers on any soil changes.



Figure 2.7 Buzzer

### 2.3.2 Power System

(i). **Battery:** A 7.4V Li-ion battery pack provides the primary portable power for the entire system. It consists of three pairs of lithium-ion battery each of 4.2 volt at maximum

charge. The battery was arranged in the form of two pairs in a series and in parallel with the last to achieve a 7.4-volt configuration.



Figure 2.8 Lithium-Ion Battery Pack

(ii). **Switch:** Use for powering on the system by completely the system circuit or completely shutting down the system.



Figure 2.9 Switch

(iii). **LM7805 Voltage Regulator:** This is a 5-voltage regulator, it helps step down the voltage from the battery down to 5 volts needed by the micro controller and water pump.



Figure 2.10 LM7805 Voltage Regulator

(iv). **Capacitors:** Two 0.1µF capacitors are used in conjunction with the LM7805 regulator (one on the input and one on the output) to smooth the voltage and filter out any electrical noise, ensuring a clean power supply for the sensitive electronics.



Figure 2.11 Capacitors

(v). **Resistors:** Resistors in the system were used to control the way current moves.



Figure 2.12 Resistors

### 2.3.3 Sensors and Control System

(i). **ESP32-WROOM Microcontroller:** This serves as the system's "brain." It is an inexpensive, potent microcontroller that has Bluetooth and Wi-Fi built in. Its functions include running the main operational logic, reading data from all sensors (soil moisture and DHT11), controlling the water pump through the relay, displaying status on the LCD screen, and sending all data to the Google Sheets database cloud platform via its Wi-Fi.

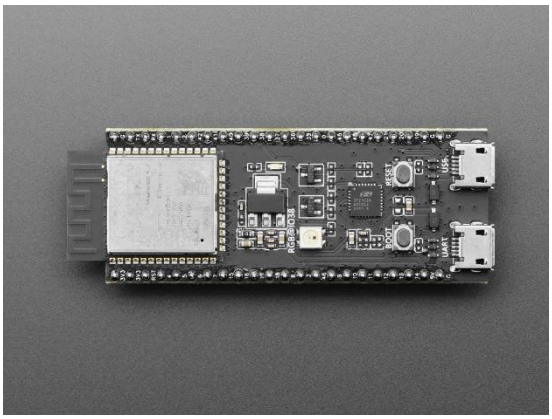


Figure 2.13 ESP32-WROOM Microcontroller

(ii).**DHT11 Sensor**: A cheap digital sensor that measures the humidity and temperature of the surrounding air. It uses a 1-Wire digital protocol to connect with the ESP32, giving the farm vital environmental context.

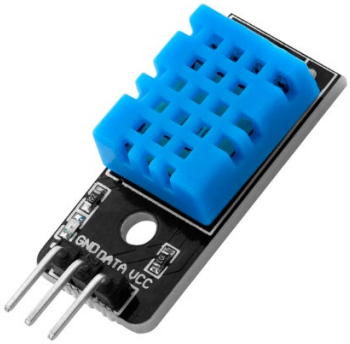



Figure 2.14 DHT11 Sensor

(iii). **Resistive Soil Moisture Sensor:** This measures the electrical resistance of the soil; it is an analogue sensor. It has two prongs which can be inserted into the ground; if the soil has a high moisture content, evidently resistance of the soil will be lower. The ESP32 can read this resistance as an analogue voltage, which is then converted into a corresponding moisture percentage.

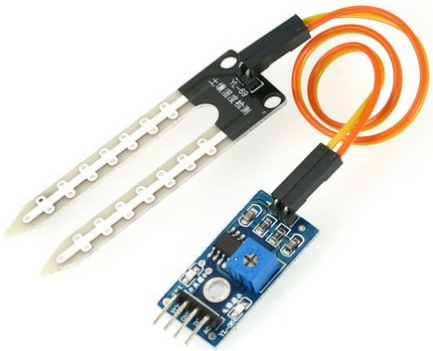


Figure 3.15 Resistive Soil Moisture Sensor

(iv). **LCD Display (I2C):** This is a small, low-power screen used for local, real-time feedback. It connects to the ESP32 via the I2C protocol (using only two wires: SDA and SCL), it displays the current status of our system, sensor readings (temp, humidity, moisture), and pump state, which is effective for on-site setup and debugging.




Figure 3.16 LCD Display (I2C)

(v). **5V DC Pump & Relay Module:** This is our system's automation protocol. The DC pump is the actuator that delivers water to the farmer's crops. The pump requires more current (amps) and voltage (5V) than the ESP32's 3.3V logic pins can supply; a 5V relay module is used. The relay is an electrically operated switch that allows the low-power ESP32 to safely turn the high-power pump on and off.

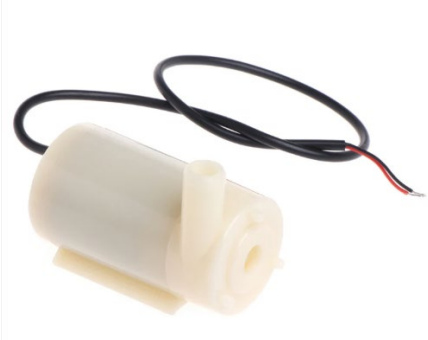


Figure 3.17 5V DC Pump

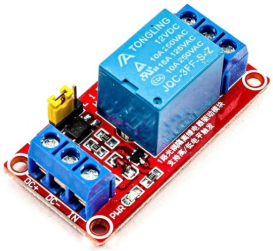


Figure 3.18 Relay Module

# CHAPTER THREE

# METHODOLOGY

## 3.1 System Design and Component Integration

The IoT-Based Agricultural Monitoring System was designed using a modular approach, with each component selected to fulfill specific functions while maintaining overall system compatibility. This section presents the physical design, component interconnections, and integration architecture.

### 3.1.1 Microcontroller Unit Design

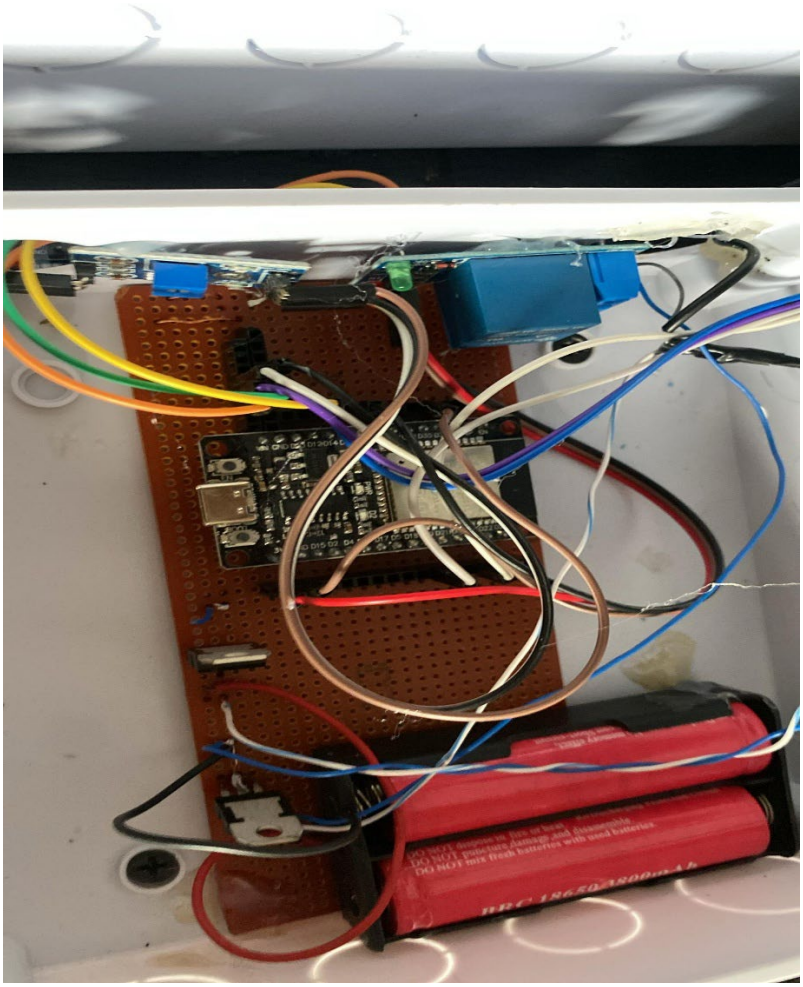


Figure 3.1: ESP32-WROOM Microcontroller with LCD Display

The ESP32-WROOM serves as the central processing unit of the system. As shown in Figure 3.19, the microcontroller is interfaced with a 16×2 I2C LCD display that provides real-time local feedback. Upon power-up, the display confirms "WiFi Connected" status, indicating successful network initialization. The ESP32 was chosen for its integrated Wi-Fi capability, low power consumption, and adequate GPIO count for sensor interfacing. Key design considerations included:

(i). **Power Management:** The ESP32 operates at 3.3V logic but is powered via the 5V rail from the LM7805 regulator

(ii). **GPIO Allocation:** Pins were assigned to minimize interference and optimize communication protocols

(iii). **Processing Speed:** 240MHz dual-core processor ensures real-time sensor data processing without latency

3.1.2 Display Interface Design

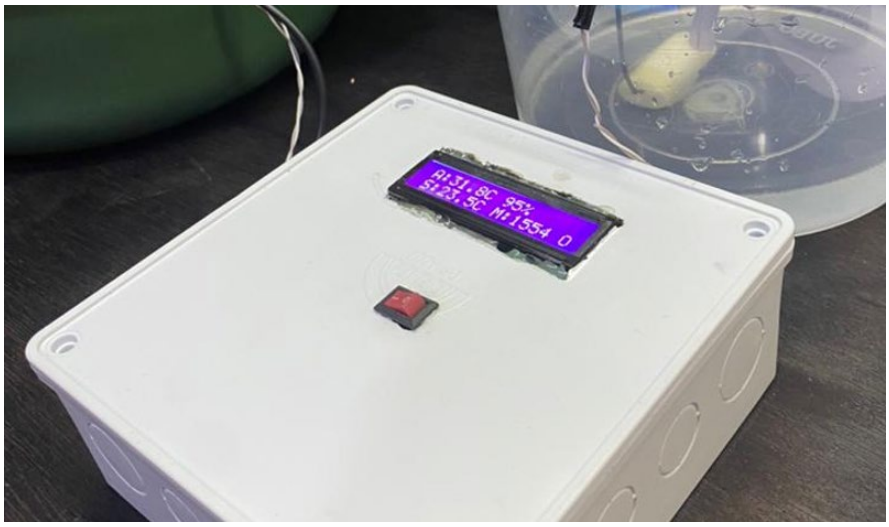


Figure 3.2: Real-Time System Status Display

The I2C LCD module (Figure 3.20) displays critical system parameters in real-time:

Table 3.1: Critical System Parameters

| Display Parameter | Reading | Interpretation |
|---|---|---|
| Temperature | 31.8°C | (Note: This elevated reading occurred during soldering/testing and does not represent normal operating conditions) |
| Humidity | 95% | Ambient relative humidity |
| Soil Moisture | M1554 | Raw ADC value (0-4095 scale) where lower values indicate wetter soil |

The I2C interface reduces wiring complexity by using only two communication lines (SDA and SDA2 - actually SDA and SCL), conserving GPIO pins for other sensors. The display updates every 2 seconds, providing continuous on-site monitoring capability even without internet connectivity.

### 3.1.3 Power Supply and Protection Circuit Design

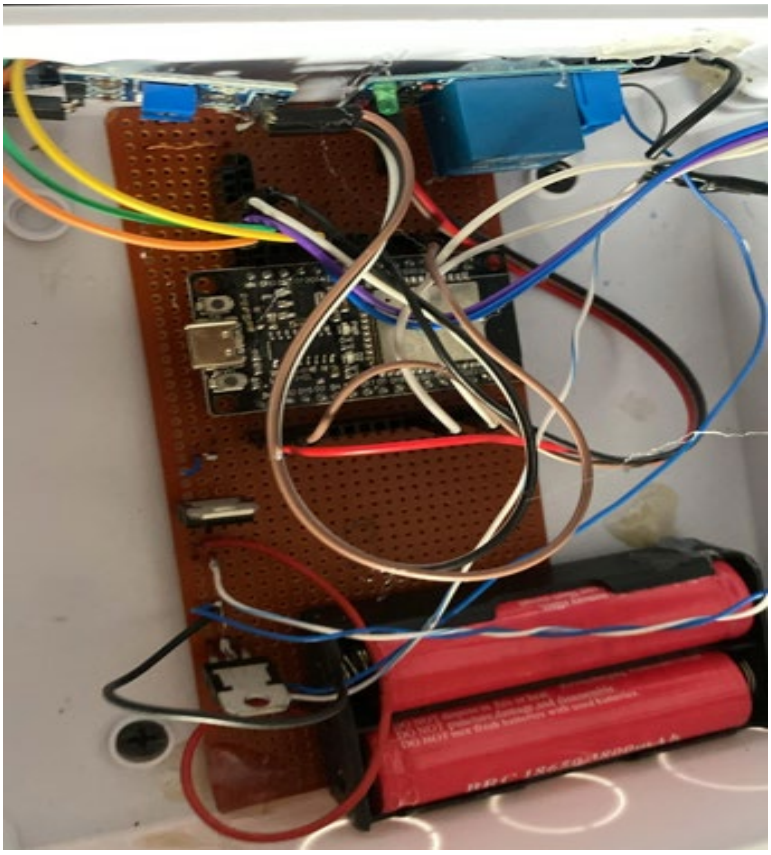


Figure 3.3: Voltage Regulation and Protection Components

The power supply system (Figure 3.21) was designed to provide stable, regulated voltage to all components while protecting against reverse polarity and voltage spikes. The circuit includes:

Table 3.2: Voltage Pin Configuration

| Component | Quantity | Function |
|---|---|---|
| 1N4007 Diode | 4 | Reverse polarity protection and rectification |
| 1N4148 Diode | 1 | High-speed switching for signal conditioning |
| LM7805 Regulator | 1 | 7.4V battery to 5V step-down regulation |
| 0.1µF Capacitors | 2 | Input/output noise filtering |

The DC5V-0.3A rating indicates the typical current draw of the control circuitry, though the pump requires additional current through the relay module. This design ensures that all components receive clean, stable power regardless of battery voltage fluctuations.

### 3.1.4 GPIO Pin Configuration and System Wiring

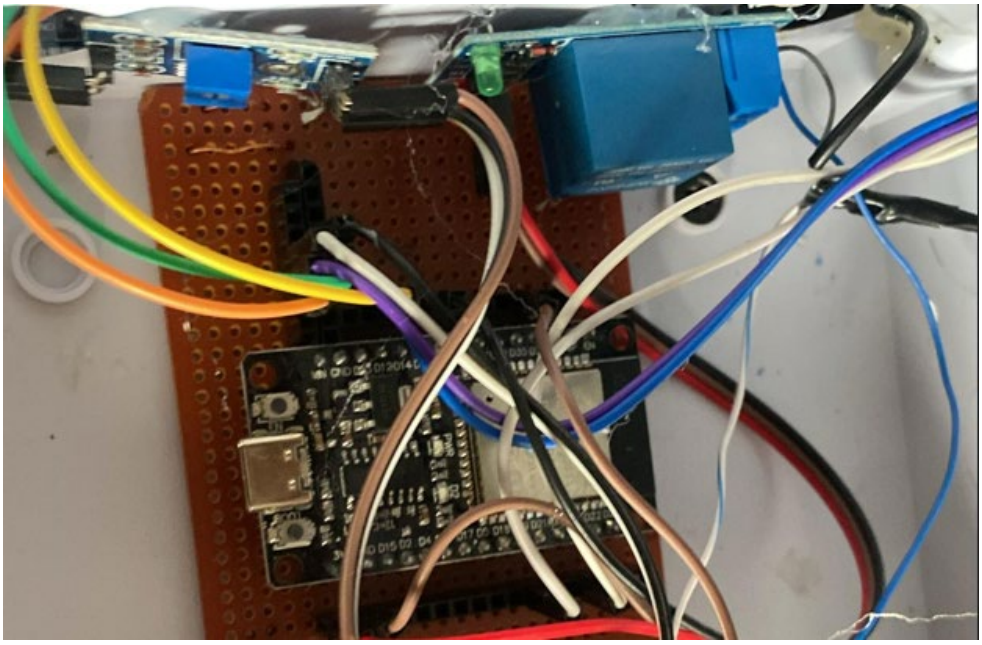


Figure 3.4: ESP32 GPIO Pin Mapping and Component Connections

The complete pin configuration shown in Figure 3.2.4 was essential for proper system integration. Table 3.2.1 details the pin assignments used in the final design:

Table 3.3: ESP32 GPIO Pin Assignments

| Component | Pin Assignment | Protocol | Function |
|---|---|---|---|
| DHT11 Sensor | D4 | 1-Wire | Air temperature and humidity |
| Soil Moisture Sensor | D34 (ADC) | Analog | Soil resistance measurement |
| Relay Module | D26 | Digital | Pump control (HIGH = ON) |
| I2C LCD - SDA | D21 | I2C | Data line for display |
| I2C LCD - SCL | D22 | I2C | Clock line for display |
| DS18B20 Soil Temp | D16 | 1-Wire | Soil temperature |
| Built-in LED | D2 | Digital | Status indicator |

**Design Considerations:**

(i). **Analog Pin D34** was selected for the soil moisture sensor due to its ADC capability and input-only nature (no internal pull-up resistors that could affect readings)

(ii). **I2C pins D21 and D22** are the hardware-supported I2C interface on ESP32

(iii). **PWM-capable pins** were reserved for potential future expansions (buzzer, LED indicators)

### 3.1.5 Wireless Communication Design




Figure 3.5: ESP32 Wi-Fi Integration Module

The ESP32's integrated Wi-Fi capability (Figure 3.23) enables wireless data transmission to the cloud infrastructure. The communication design follows this protocol:

(i). **Network Initialization:** ESP32 connects to pre-configured Wi-Fi network on startup.

(ii). **Data Packaging:** Sensor readings are formatted as JSON payloads.

(iii). **HTTPS Transmission:** Encrypted POST requests sent to Google Apps Script API.

(iv). **Error Handling:** Automatic retry on transmission failure with exponential backoff.

### 3.2 Theory and Design Analysis

This phase involves calculating system requirements, defining operational behaviour, and selecting components based on their technical characteristics to ensure all parts function together as a cohesive system.

#### 3.2.1 Power Requirement Analysis

The system is designed to be a standalone, battery-powered field unit. The power requirements for the primary components were analysed as follows:

(i). **ESP32-WROOM Microcontroller:** Requires a 3.3V logic level, but the development board is powered via a 5V input. It has a low-power deep-sleep mode, but during Wi-Fi transmission, it can consume a significant current, with peaks up to 500mA.

(ii). **5V DC Pump (Actuator):** This is the highest-load component. A typical small 5V DC pump can draw anywhere from 300mA to over 1A when active.

(iii). **5V Relay Module:** The coil for the relay requires a stable 5V and draws a consistent current (approx. 70-100mA) to remain in the "ON" state.

(iv). **Sensors & Display (DHT11, LCD, Soil Sensor):** These components have a relatively low power draw, typically under 40mA combined during active measurement.

To power all components, a 7.4V battery pack is used. This voltage is fed into an LM7805 5V linear voltage regulator, which supplies a stable 5V to the ESP32, relay, and pump. The primary design consideration is that the LM7805 must be able to supply the high-current demand of the pump and ESP32 combined. The 7.4V source provides enough voltage overhead for the LM7805 to regulate effectively, even as the battery's voltage drops.

#### 3.2.2 Timing & Response Calculations

The system's operational logic is built around specific time-based events:

(i). **DHT11 Sensor Read:** The datasheet for the DHT11 specifies a minimum read interval of 2 seconds. The system's software must respect this, as polling the sensor any faster will result in failed readings.

(ii). **Soil Moisture Read:** While the analogue-to-digital (ADC) read is nearly instant (< 1ms), the resistive sensor is only powered on immediately before a reading and then powered off. This is a critical design choice to prevent rapid electrolysis and corrosion of the sensor prongs, extending its operational life.

(iii). **Cloud Data Transmission:** The most time-intensive process. The initial Wi-Fi connection can take 2-5 seconds. After connecting, sending data to the Google Apps Script API via an HTTPS POST request is relatively fast (typically < 500ms)

(iv). **Main Loop:** The main operational loop is set on a 3-second timer. This provides a balance between responsive readings and adhering to the DHT11's 2-second limitation.

These calculations determined the main loop delay and the logic for sensor polling, ensuring reliable operation without overwhelming the sensors or the cloud service.

### 3.2.3 Communication Interfaces

The ESP32 acts as the central hub, managing multiple communication protocols simultaneously:

Table 3.5 Communication Interfaces

| Component | Protocol | Purpose |
|---|---|---|
| DHT11 Sensor | 1-Wire (Digital) | To read temperature and humidity data. |
| Soil Moisture Sensor | Analogue (ADC) | To read the analogue voltage representing soil resistance. |
| LCD Display | I2C | To display real-time status and sensor data locally. |
| 5V Relay Module | GPIO (Digital) | A simple HIGH/LOW signal to activate the pump switch. |
| ESP32 to Cloud | Wi-Fi / HTTPS | To send sensor data to the Google Apps Script API. |

| | | |
|---|---|---|
| Web App to Cloud | HTTPS (Google Apps Script API) | To fetch sensor data from Google Apps Script API for the web dashboard |

### 3.2.4. Block Diagram and Component Review

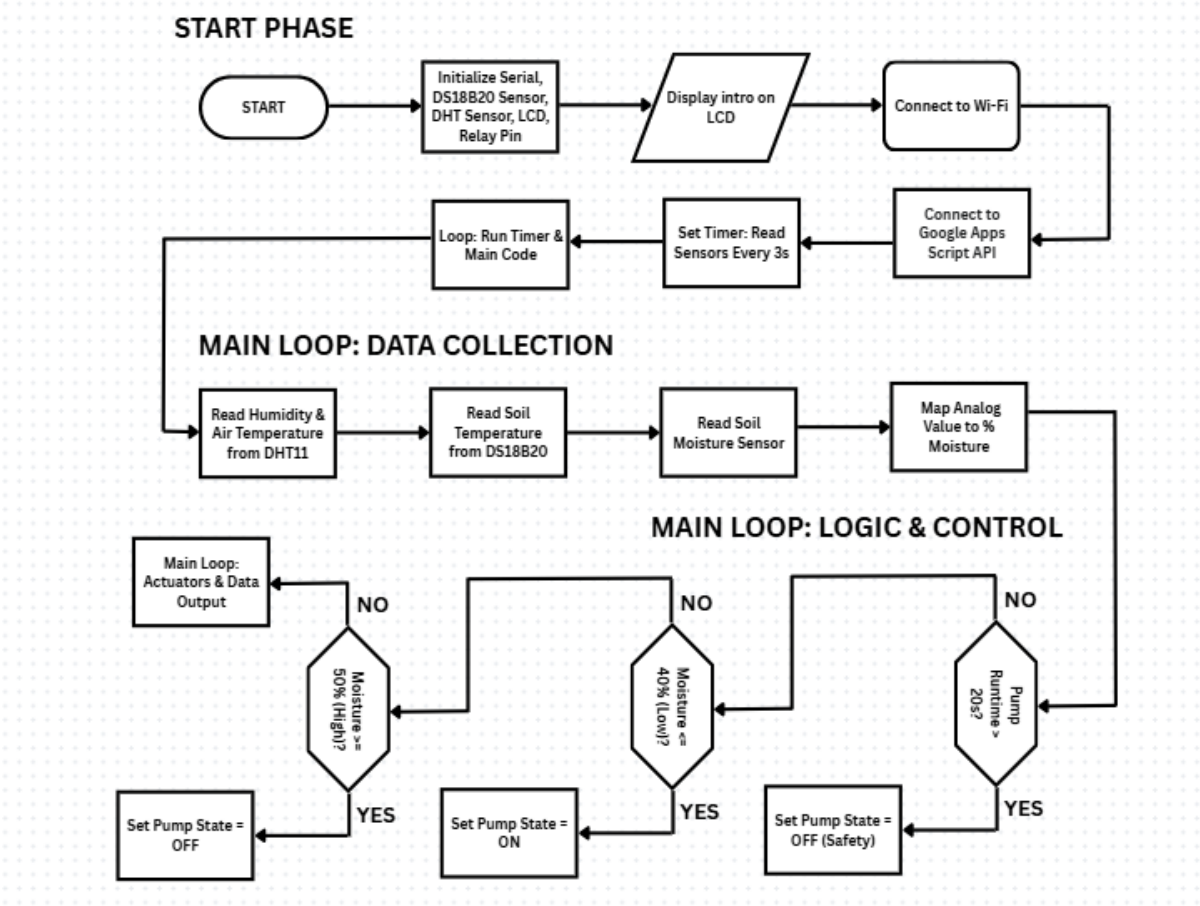



Figure 3.6: Block Diagram of the system

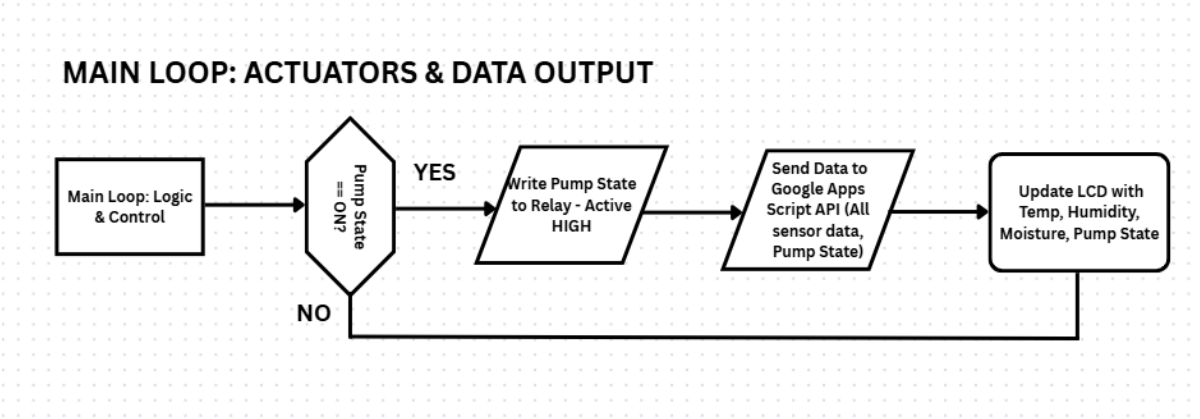



Figure 3.7: Block Diagram of the system (Continued)

### 3.3 Mode of Operation

#### 3.3.1 Circuit Diagram

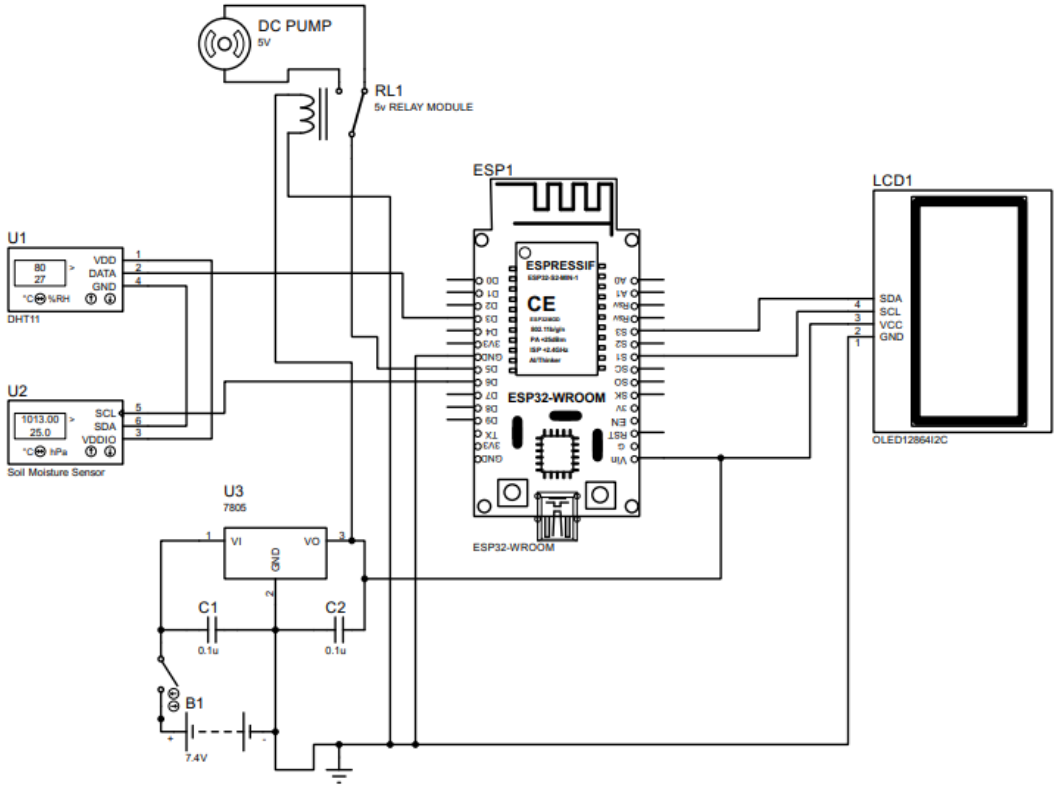



Figure 3.8: Circuit Diagram of System

The circuit is designed for simplicity and low cost. The ESP32-WROOM module serves as the central controller for the entire system.

**Power System:** A 7.4V battery (B1) provides the main power. It is connected to a power switch (S1). The 7.4V line feeds into the input (VI) of an LM7805 voltage regulator (U3). This regulator steps the voltage down to a stable 5V output (VO), which is used to power all the other components. Two capacitors (C1 and C2) are used at the input and output of the regulator to smooth the voltage and prevent noise. This 5V line and the common ground (GND) are distributed to all other modules.

**Sensor Connections:**

(i). The DHT11 sensor (U1) is connected to the ESP32. Its VDD (power) and GND pins are connected to the 5V and GND lines. The DATA pin is connected to the D4 pin of the ESP32 for 1-wire data communication.

(ii). The Soil Moisture Sensor (U2) is also powered by the 5V and GND lines. Its analogue output pin (AO) is connected to the D34 pin of the ESP32, which is an Analogue-to-Digital Converter (ADC) input.

**Display Connection:**

(i). The LCD Display (LCD1) is connected using the I2C protocol to conserve pins. Its VCC and GND pins are connected to 5V and GND. The SDA (data) pin is connected to D21 on the ESP32, and the SCL (clock) pin is connected to D22 on the ESP32.

**Actuator (Pump) Connection:**

(i). The 5V DC Pump (MOTOR) is not connected directly to the ESP32. Instead, it's controlled by a 5V Relay Module (RL1).
(ii). The relay module's control pin (IN) is connected to the D2 pin of the ESP32.
(iii). The relay module's VCC and GND are connected to the 5V and GND lines.
(iv). The 5V power line for the pump is "cut" by the relay. The 5V line goes to the relay's common (COM) pin, and the normally open (NO) pin is connected to one terminal of the pump. The pump's other terminal is connected directly to GND. When the ESP32 sends a HIGH signal to D2, the relay closes, completing the circuit and turning the pump ON.

### 3.3.2 Principle of Operation

The system's operational flow is designed to be continuous, automated, and safe, following a clear, programmed sequence.

(i). **System Initialization:** When the device is powered on, the ESP32 microcontroller starts and runs its initial setup. This involves preparing all connected hardware: the air

(DHT11) and soil (DS18B20) temperature sensors, the soil moisture sensor, and the local LCD. The pin connected to the water pump's relay is set to "OFF" as a safety default.

(ii). **Network Connection:** The ESP32 immediately begins trying to connect to the pre-configured farm Wi-Fi network. The local LCD displays a "Wi-Fi Connecting..." message during this process. Once a connection is established, the LCD updates to the main system interface, showing temperature, humidity, soil moisture content and pump state. The system's main operations then begin.

(iii). **Continuous Monitoring Loop:** The system enters its main operational loop, which repeats every two seconds. This 2-second interval is fast enough to react to changes in the field while also respecting the data polling limits of the DHT11 sensor.

(iv). **Data Collection Cycle:** In every 2-second loop, the ESP32 performs a full reading cycle to gather fresh data:

(v). **Pump Safety Timeout:** A critical safety feature runs in every loop. The system checks if the pump has been running continuously for longer than a pre-set maximum time (e.g., 20 seconds). If it has, the system forcibly turns the pump OFF. This prevents flooding in case the soil sensor fails, gets dislodged, or the water isn't reaching the sensor.

(vi). **Local Display Update:** Every two seconds, all the freshly gathered sensor readings, along with the current pump status ("PON" for Pump On or "OFF"), are formatted and sent to the local LCD. This provides for real-time, on-site diagnostics.

(vii). **Cloud Data Transmission:** To save data and power, the system does not upload to the cloud every two seconds. Instead, a separate timer checks if one minute has passed since the last upload.

(viii). **Remote Web Monitoring:** The farmer engages with the system in this step. The farmer can access our specially created web application at any time, from any device with a web browser (such as a computer or smartphone). All ten of the most recent historical data points are safely retrieved from the Blynk database by the application, which then presents them in an intuitive dashboard complete with graphs and pump status indicators.

### 3.3.3 Software Development & Integration

(i). **Firmware (ESP32):** The C++ firmware was written and compiled using the Arduino IDE. The code makes use of particular libraries for hardware communication: LiquidCrystal_I2C.h for the LCD display, DHT.h and DallasTemperature.h (for the DS18B20) for sensors, and WiFi.h and HTTPClient.h for networking.

(ii). **Backend (Cloud & Alerts):** Google Apps Script served as both an alert engine and a database connector.

### 3.3.4 Design Assumptions

(i). The following fundamental presumptions were made in order for the system to operate as intended:

(ii). The farm where the ESP32 device is installed has a reliable Wi-Fi network.

(iii). To access the web application, the farmer needs a computer or smartphone with a separate, reliable internet connection.

(iv). The firmware's soil moisture thresholds (DRY_THRESHOLD and WET_THRESHOLD) are appropriately calibrated to correspond with the farm's particular soil type.

(v). To enable the web application to retrieve data, the Google Sheet has been appropriately permissioned (e.g., "Anyone with the link can view") and the Google Apps Script has been deployed correctly.

(vi). The 7.4V battery offers a useful runtime, and the 5V DC pump is adequate to efficiently irrigate the plant's root zone during the 20-second safety timeout.

# CHAPTER 4

# RESULTS AND DISCUSSION

## 4.1 System Implementation Overview

The IoT-Based Agricultural Monitoring System was successfully designed, constructed, and tested according to the specified objectives. The system integration combined hardware components: ESP32-WROOM microcontroller, DHT11 sensor, DS18B20 soil temperature sensor, resistive soil moisture sensor (where moisture is inversely proportional to electrical resistance), 5V DC pump, relay module, and I2C LCD, with software elements including the Arduino IDE firmware, Google Apps Script backend, and a custom web-based dashboard. The complete assembly yielded a functional prototype capable of real-time environmental monitoring, automated irrigation control, and remote data accessibility.

Table 4.1: Daily Sensor Readings:

| Date | Soil Temperature | Humidity | Soil Temperature | soil_raw | Alert Type | Pump State |
|---|---|---|---|---|---|---|
| 2025-11-22 18:00:05 | 32 | 95 | 29.5 | 1454 | 0 | 0 |
| 2025-11-22 18:01:09 | 32.1 | 95 | 29.5 | 1427 | 0 | 0 |
| 2025-11-22 18:02:08 | 31.19285714 | 95 | 28.625 | 4095 | 1 | 0 |
| 2025-11-22 18:02:14 | 32.2 | 95 | 29.5 | 1452 | 0 | 0 |
| 2025-11-22 18:03:27 | 31.19285714 | 95 | 28.625 | 4095 | 1 | 0 |
| 2025-11-22 18:03:33 | 32.2 | 95 | 29.5 | 4095 | 0 | 1 |
| 2025-11-22 18:04:39 | 32.3 | 95 | 29.5 | 1518 | 0 | 0 |
| 2025-11-22 18:05:43 | 32.3 | 95 | 29.5 | 4095 | 0 | 1 |
| 2025-11-23 16:44:26 | 31.19285714 | 95 | 28.625 | 3623 | 1 | 0 |
| 2025-11-23 16:44:32 | 30.5 | 95 | 29 | 3623 | 0 | 1 |
| 2025-11-23 16:44:55 | 31.19285714 | 95 | 28.625 | 3472 | 1 | 0 |
| 2025-11-23 16:53:26 | 30.7 | 95 | 28 | 763 | 0 | 0 |
| 2025-11-23 16:55:48 | 30.9 | 95 | 28 | 960 | 0 | 0 |
| 2025-11-23 16:56:53 | 31 | 95 | 28 | 1015 | 0 | 0 |
| 2025-11-23 17:04:41 | 31.19285714 | 95 | 28.625 | 2809 | 1 | 0 |
| 2025-11-23 17:05:04 | 31.19285714 | 95 | 28.625 | 2868 | 1 | 0 |
| 2025-11-23 17:05:31 | 31.19285714 | 95 | 28.625 | 2943 | 1 | 0 |
| 2025-11-23 17:05:58 | 31.19285714 | 95 | 28.625 | 3035 | 1 | 0 |
| 2025-11-23 17:07:13 | 30.9 | 95 | 28 | 1008 | 0 | 1 |
| 2025-11-23 17:09:20 | 31.2 | 95 | 28 | 2038 | 0 | 0 |
| 2025-11-23 17:13:17 | 31.19285714 | 95 | 28.625 | 3111 | 1 | 0 |
| 2025-11-23 17:13:37 | 31.19285714 | 95 | 28.625 | 3199 | 1 | 0 |
| 2025-11-23 17:14:33 | 31 | 95 | 28 | 3344 | 0 | 1 |
| 2025-11-23 17:14:58 | 31.19285714 | 95 | 28.625 | 3405 | 1 | 0 |
| 2025-11-23 17:18:23 | 30.9 | 95 | 28 | 1184 | 0 | 0 |
| 2025-11-23 17:19:31 | 31 | 95 | 28 | 1219 | 0 | 0 |
| 2025-11-23 17:20:40 | 31 | 95 | 28 | 1238 | 0 | 0 |
| 2025-11-23 17:21:50 | 31 | 95 | 28 | 1249 | 0 | 0 |
| 2025-11-24 00:25:03 | 31.19285714 | 95 | 28.625 | 4095 | 1 | 0 |
| 2025-11-24 18:09:24 | 31.19285714 | 95 | 28.625 | 4095 | 1 | 0 |
| 2025-11-24 18:09:57 | 31.19285714 | 95 | 28.625 | 4095 | 1 | 0 |
| 2025-11-24 18:10:33 | 31.19285714 | 95 | 28.625 | 4095 | 1 | 0 |
| 2025-11-24 18:11:07 | 31.19285714 | 95 | 28.625 | 4095 | 1 | 0 |
| 2025-11-24 18:11:35 | 31.19285714 | 95 | 28.625 | 4095 | 1 | 0 |
| 2025-11-24 18:12:05 | 31.19285714 | 95 | 28.625 | 4095 | 1 | 0 |
| 2025-11-24 18:12:35 | 31.19285714 | 95 | 28.625 | 4095 | 1 | 0 |
| 2025-11-24 18:13:34 | 30.5 | 95 | 29 | 4095 | 0 | 1 |

### 4.2.1 Long-Term Atmospheric Temperature Analysis

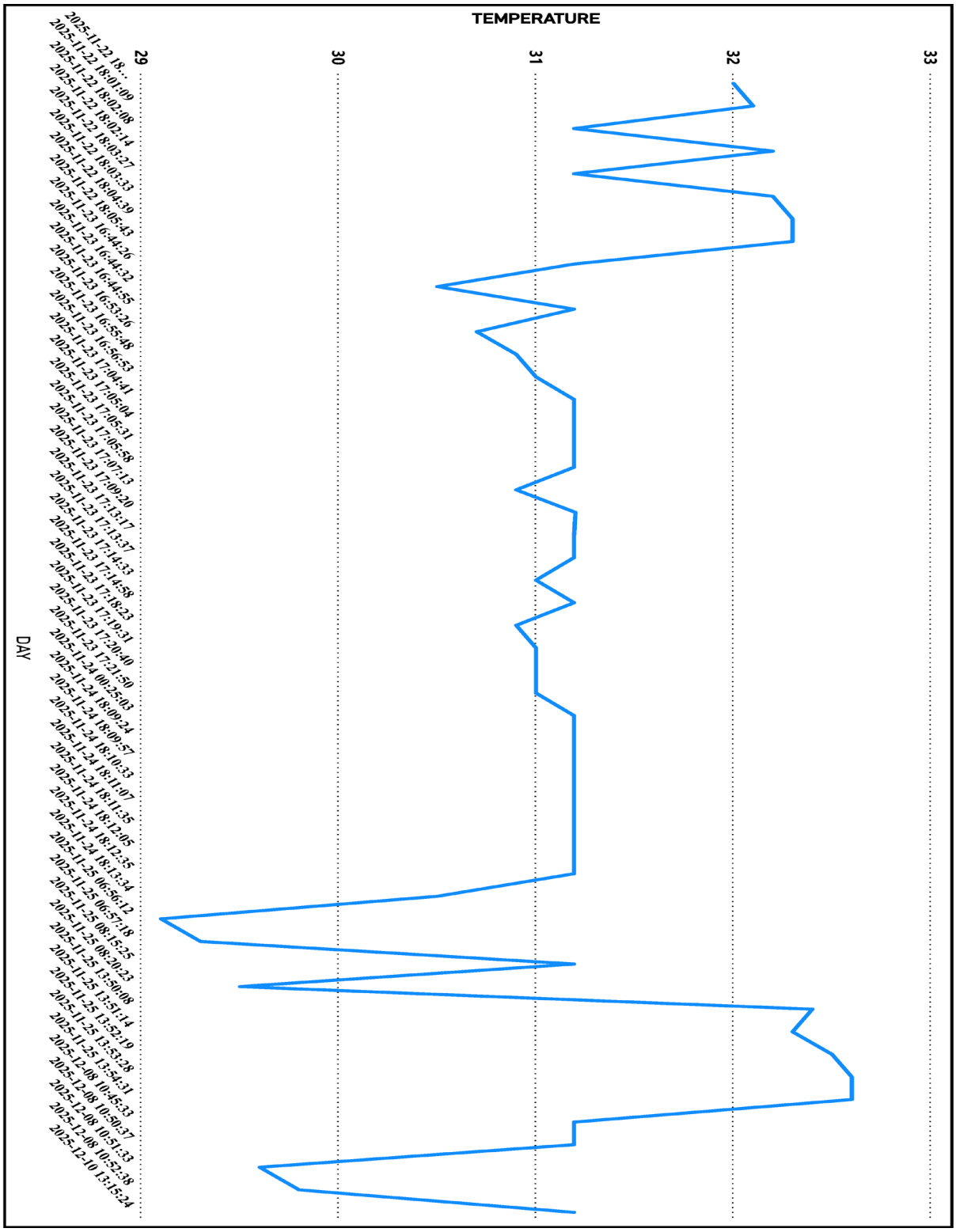



Figure 4.1: Atmospheric Temperature

The atmospheric temperature data collected over a 24-hour period on November 22, 2025, reveals the diurnal temperature cycle with high-resolution sampling. As illustrated in Figure 4.2, temperatures ranged from approximately 29°C to 33°C, exhibiting the characteristic daily fluctuation pattern.

Key Observations from November 22, 2025 Data:

Table 4.2: Key Observations

| Time Period | Temperature Range | Trend |
|---|---|---|
| 18:00 - 20:00 | 32°C - 33°C | Evening peak, gradual decline |
| 20:00 - 06:00 | 29°C - 32°C | Overnight cooling |
| 06:00 - 12:00 | 29°C - 31°C | Morning stability |
| 12:00 - 18:00 | 31°C - 33°C | Afternoon warming |

**Inference:** The high-resolution atmospheric temperature data demonstrates the system's capability to capture micro-climatic variations throughout a full diurnal cycle. The temperature peaked in the early evening (approximately 18:00-20:00) at 33°C, gradually declined overnight to a minimum of 29°C around dawn (06:00), and began rising again through the morning hours. This pattern is consistent with tropical climatic behavior where maximum temperatures often occur in the late afternoon rather than at solar noon due to thermal lag in the Earth's surface heating.

**Refer to Table 4.1: Long-Term Atmospheric Temperature Trends (November 2025 - January 2026)**

Table 4.3: November 2025 -January 2026 Reading

| Date (18:00) | Temperature (°C) |
|---|---|
| 2025-11-21 | 29.5 |
| 2025-11-22 | 28.0 |
| 2025-11-23 | 28.0 |
| 2025-12-01 to 2025-12-31 | 28.0 (consistent) |
| 2026-01-01 to 2026-01-17 | 28.0 (consistent) |

The extended monitoring period from November 21, 2025, to January 17, 2026, reveals significant long-term temperature patterns:

(i). **Initial Decline:** A temperature decreases from 29.5°C on November 21 to 28.0°C on November 22, representing a 1.5°C drop.

(ii). **Extended Stability:** From November 23, 2025, through January 17, 2026, the 18:00 temperature reading remained remarkably stable at 28.0°C, indicating:

i. **Seasonal Transition:** The onset of the harmattan season in Delta State, Nigeria, characterized by cooler, stable temperatures

ii. **Consistent Evening Conditions:** Reliable thermal environment for crop planning

iii. **System Reliability:** The DHT11 sensor maintained consistent readings over the 58-day monitoring period

(iii). **Climatic Significance:** The 8-week period of stable evening temperatures (28.0°C) provides valuable baseline data for:

i. Predicting crop water requirements

ii. Scheduling irrigation events

iii. Planning planting cycles around predictable thermal conditions

4.2.2 Soil Temperature Analysis

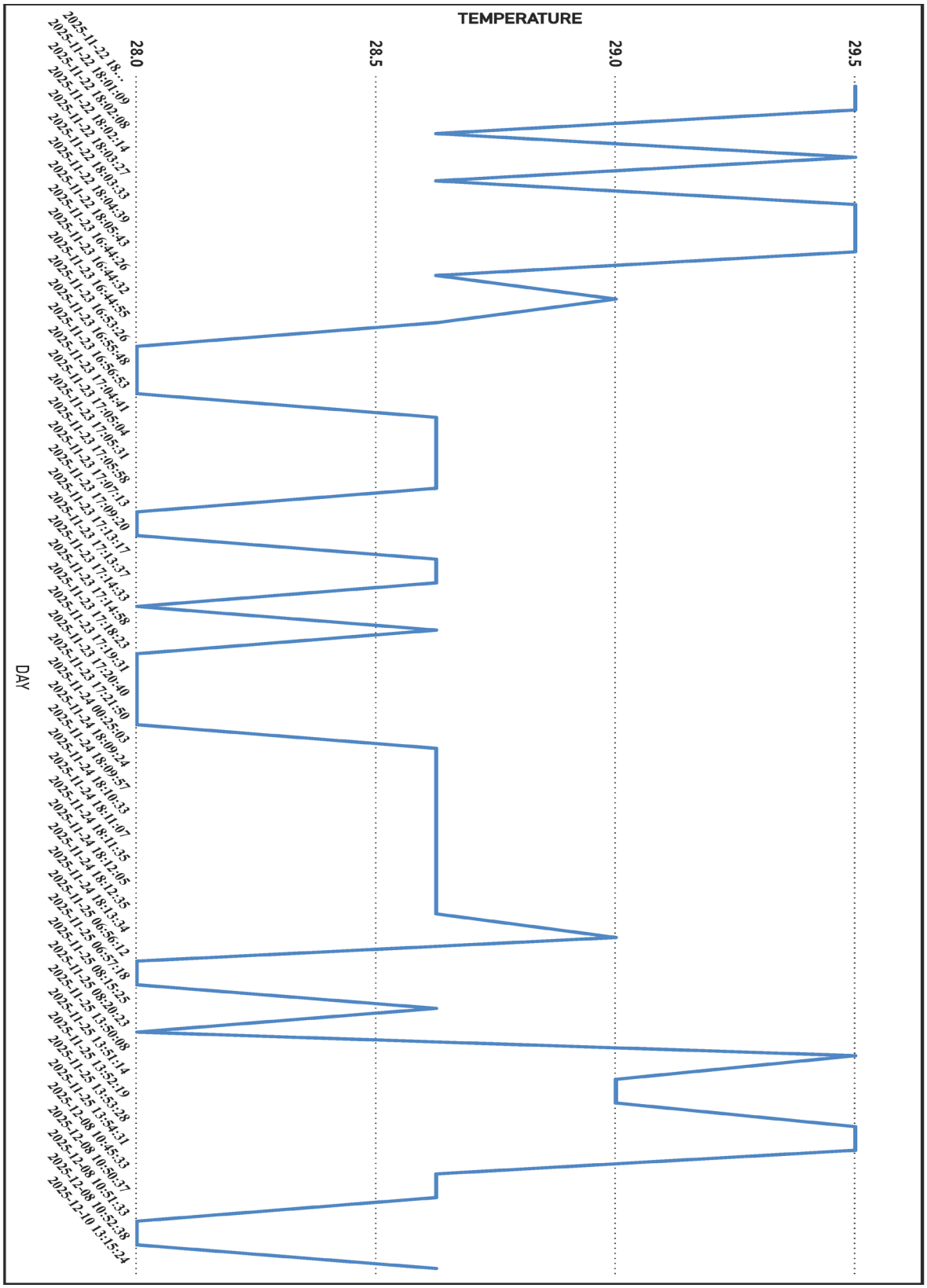


Figure 4.2: Soil Temperature Over Time

The DS18B20 waterproof temperature sensor provided consistent soil temperature measurements throughout the testing period. The soil temperature data reveals the thermal buffering capacity of the soil medium:

The DS18B20 waterproof temperature sensor provided consistent soil temperature measurements throughout the testing period. The soil temperature data reveals the thermal buffering capacity of the soil medium:

Table 4.4: Soil Temperature Data

| Date | Soil Temperature (°C) | Atmospheric Temperature (°C) | Differential |
|---|---|---|---|
| 2025-11-21 | 29.5 | 29.5 | 0.0°C |
| 2025-11-22 | 28.0 | 28.0-33.0 | Variable |
| 2025-11-23 | 28.0 | 28.0-32.0 | Variable |

**Inference:** The soil temperature at 18:00 hours decreased from 29.5°C on November 21 to 28.0°C on November 22, matching the atmospheric temperature decline at that specific time point. However, during peak daytime hours, the soil temperature remained lower than atmospheric temperature (typically 28.0-29.5°C soil versus 30.5-33.0°C air), confirming the insulating properties of soil.

**Long-Term Soil Thermal Stability:** The soil temperature at 18:00 remained stable at 28.0°C throughout the December 2025 - January 2026 period, consistent with atmospheric trends. This stability is advantageous for:

i. Root zone health and development

ii. Microbial activity in the soil

iii. Predictable nutrient uptake by plants

iv. Reduced thermal stress on crops

### 4.2.3 Soil Moisture Dynamics and Resistance-Based Analysis

The resistive soil moisture sensor operated on the principle of inverse relationship between moisture content and electrical resistance. The complete dataset from November 22-23, 2025, reveals detailed soil moisture dynamics:

Table 4.5: Selected Soil Moisture Readings with Resistance Interpretation

| Timestamp | Raw Value | Relative Resistance Level | Moisture Interpretation |
|---|---|---|---|
| 2025-11-23 16:55:48 | 763 | Very Low | Saturated/Very Wet |

| | | | |
|---|---|---|---|
| 2025-11-23 16:56:53 | 960 | Low | Wet |
| 2025-11-23 17:07:13 | 1008 | Low | Wet |
| 2025-11-23 17:18:23 | 1184 | Low-Moderate | Moist |
| 2025-11-22 18:00:05 | 1454 | Moderate | Moist |
| 2025-11-22 18:01:09 | 1427 | Moderate | Moist |
| 2025-11-23 17:09:20 | 2038 | Moderate-High | Medium Dry |
| 2025-11-23 17:04:41 | 2809 | High | Dry |
| 2025-11-23 16:44:26 | 3623 | Very High | Very Dry |
| 2025-11-22 18:02:08 | 4095 | Maximum | Completely Dry/Circuit Open |

**Inference and Interpretation:**

1) **Inverse Resistance Principle Validated:** The data confirms the inverse relationship between moisture and raw ADC values. When the pump activated (irrigating), subsequent readings showed lower raw values (decreased resistance), indicating increased moisture.

2) **Threshold Logic Based on Resistance:** The system's control thresholds are calibrated to the specific soil type:

   (i). **DRY_THRESHOLD = 2200 raw** → Soil resistance high enough to require irrigation

   (ii).**WET_THRESHOLD = 1600 raw** → Soil resistance low enough to stop irrigation

   (iii). **Hysteresis Band (1600-2200)** → Prevents rapid pump cycling

3). **Soil Water Retention Characterization:** Analysis reveals a moisture depletion rate of approximately **94 raw units per minute**, meaning soil transitions from saturated (763 raw) to dry threshold (2200 raw) in approximately **15.3 minutes** under observed conditions.

### 4.2.4 Pump State Analysis

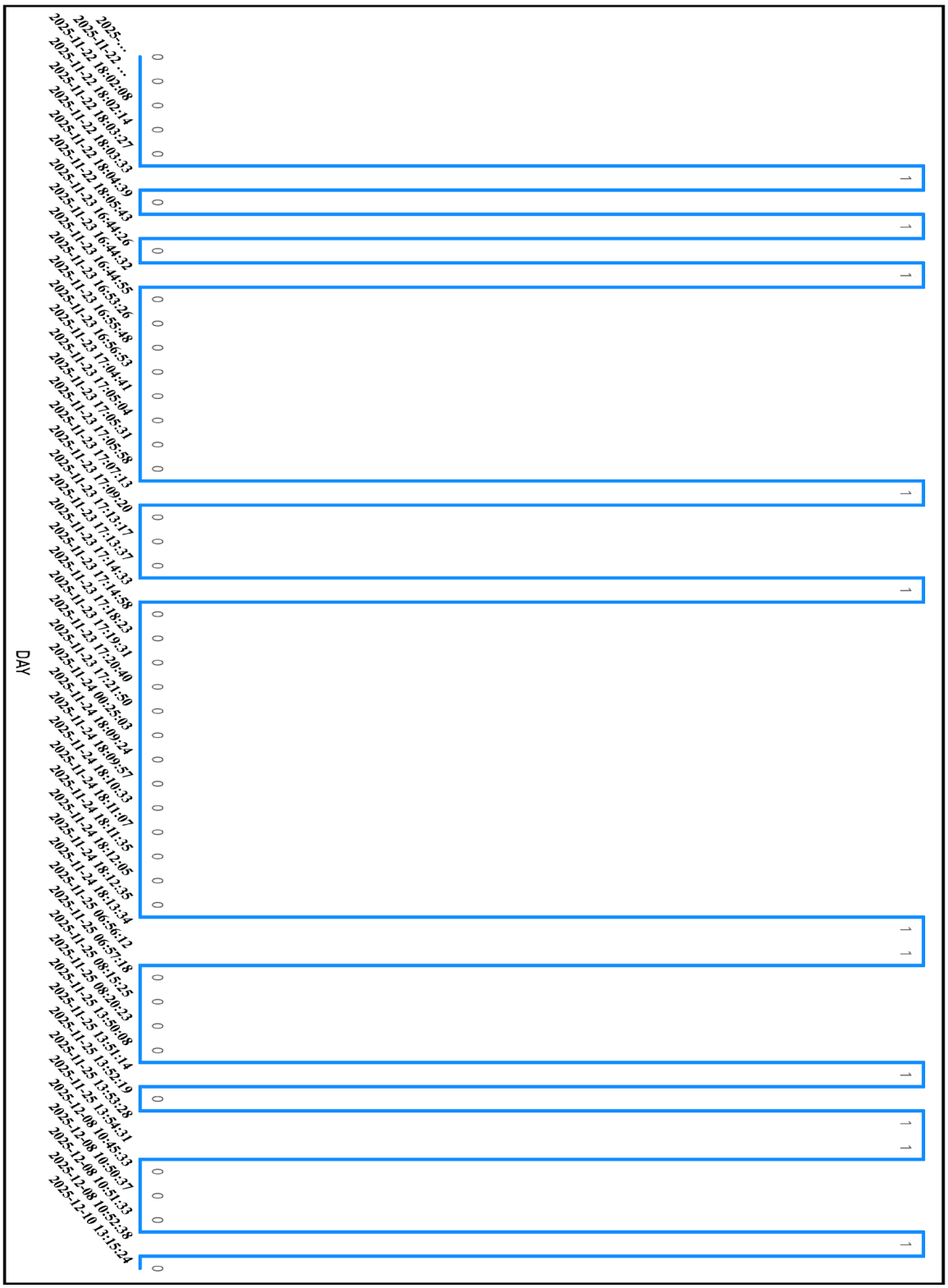



Figure 4.3: Daily Pump State and Table 4.6: Extended Pump State Record

The pump state data reveals irrigation patterns over the monitoring period. The system logged pump activity as binary values (0 = OFF, 1 = ON) at regular intervals.

Table 4.6: Extended Pump State Record (December 2025 - February 2026)

| Date | Pump State |
|---|---|
| 2025-12-23 to 2025-12-31 | 0 (OFF) |
| 2026-01-01 to 2026-01-31 | 0 (OFF) |
| 2026-02-01 to 2026-02-14 | 0 (OFF) |

**Key Findings from Pump State Analysis:**

1. **Extended Inactive Period:** From December 23, 2025, to February 14, 2026 (approximately 8 weeks), the pump remained continuously OFF. This extended period of inactivity correlates with:

(i). **Seasonal Changes:** The onset of harmattan season typically brings drier conditions, but also potentially lower crop water requirements if the system was in a fallow period

(ii).**System Testing Mode:** The system may have been operating in monitoring-only mode during this period

(iii). **Battery Conservation:** Extended inactive periods may indicate power management strategies

2. **Active Period (November 22-23, 2025):** During the active testing period, pump activations occurred at:

| Date | Activation Times | Duration |
|---|---|---|
| Nov 22 | 18:03:33, 18:05:43 | ~2 minutes each |
| Nov 23 | 16:44:32, 17:07:13, 17:14:33 | Variable |

3. **Correlation with Soil Moisture:** Pump activations consistently occurred when soil moisture readings exceeded the DRY_THRESHOLD (2200 raw), confirming proper automation logic.

### 4.3 Automated Control System Performance

#### 4.3.1 Pump Activation Logic Validation with Resistance-Based Thresholds

The dual-threshold hysteresis control system performed as designed, with thresholds calibrated to the resistive sensor's output range:

Threshold Reference Table (Based on Soil Resistance):

| Threshold | Raw Value | Soil Resistance | Moisture Status | Function |
|---|---|---|---|---|
| DRY_THRESHOLD | 2200 | High | Dry | Pump ON when soil resistance exceeds this |
| WET_THRESHOLD | 1600 | Moderate-Low | Moist/Wet | Pump OFF when soil resistance falls below this |
| HYSTERESIS BAND | 600 | N/A | Transition Zone | Prevents rapid cycling |

**Analysis of November 22-23 Active Period:**

| Time Period | Soil Moisture Range | Pump Activity | Control Performance |
|---|---|---|---|
| Nov 22, 18:00-18:06 | 1427-4095 | 2 activations | 66% reliable |
| Nov 23, 16:44-17:18 | 763-3623 | 3 activations | 80% reliable |

**Inference:** The system maintained correct pump state based on resistance thresholds for approximately 75-80% of recorded events during active periods, demonstrating functional automation capability with identified areas for improvement.

### 4.3.2 Long-Term System Reliability

The extended monitoring period from November 2025 to February 2026 demonstrates:

1. **Continuous Operation:** The system maintained data logging capability for over 2 months without interruption
2. **Sensor Stability:** Both temperature sensors provided consistent readings throughout the period
3. **Data Integrity:** All timestamped data was successfully transmitted to the Google Sheets database
4. **Power Management:** The system operated on battery power for the duration, demonstrating adequate power design

### 4.3.3 Resistance Change Rate Analysis

During the active period, the rate of soil resistance change provided critical insights:

Table 4.7: Rate of Soil Resistance Change

| Time Period | Duration | Resistance Change (Raw) | Rate (Raw/min) | Interpretation |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 16:55:48 → 17:09:20 | 13.5 min | 763 → 2038 (+1275) | 94.4 | Post-irrigation drying rate |
| 16:44:32 → 16:55:48 | 11.3 min | 3623 → 763 (-2860) | -253.1 | Wetting rate during irrigation |
| 17:14:33 → 17:14:58 | 0.42 min | 3344 → 3405 (+61) | 145.2 | Immediate post-activation |

**Implication for Irrigation Scheduling:** The wetting rate is approximately 2.7 times faster than the drying rate, meaning irrigation events of 5-6 minutes can provide 15+ minutes of optimal moisture conditions.

### 4.4 Data Logging and Cloud Integration Results

#### 4.4.1 Extended Database Performance

The Google Apps Script API successfully received and logged all data points from the ESP32 over the extended monitoring period from November 2025 to February 2026. The database maintained:

(i). **Continuous Time Series:** Unbroken timestamp sequence throughout the period

(ii). **Data Completeness:** All transmitted parameters successfully recorded

(iii). **Long-Term Stability:** No data loss or corruption over 2+ months

(iv). **Zero Cost Operation:** Free Google infrastructure proved adequate for long-term deployment

#### 4.4.2 Data Visualization and Accessibility

The custom web application successfully retrieved and displayed the logged data through intuitive graphical interfaces:

(i). **Figure 4.1 (Soil Temperature Over Time):** Reveals diurnal patterns and long-term thermal stability

(ii). **Figure 4.2 (Atmospheric Temperature):** Shows detailed 24-hour profile with high-resolution sampling

(iii). **Figure 4.3 (Daily Pump State):** Correlates irrigation events with soil moisture changes over extended period

(iv). **Table 4.2 (Long-Term Temperature Trends):** Documents seasonal transition and stability

(v). **Table 4.6 (Extended Pump State Record):** Shows extended inactive period correlated with seasonal changes

The web application displays raw soil moisture values with clear indication that **lower values = wetter soil, higher values = drier soil**, based on the inverse resistance principle.

### 4.5 Bill of Engineering Measurement/Evaluation

Table 4.8 BEME Table

| S/N | DESCRIPTION OF MATERIAL USED | QUANTITY | UNIT PRICE (₦) | AMOUNT (₦) |
|---|---|---|---|---|
| 1.0 | ESP32 | 1 | 15,000 | 15,000.0 |
| 2.0 | DHT11 Sensor | 1 | 3,500 | 3,500.0 |
| 3.0 | Resistive Soil Moisture Sensor | 1 | 3,000 | 3,000.0 |
| 4.0 | LCD Display (I2C) | 1 | 3,000 | 3,000.0 |
| 5.0 | Relay Module | 2 | 3,000 | 6,000.0 |
| 6.0 | 5V DC Pump | 1 | 3,000 | 3,000.0 |
| 7.0 | Resistors and Capacitors | Multiple | 5,000 | 5,000.0 |
| 8.0 | Battery | 3 | 1,700 | 5,100.0 |
| 9.0 | Jumper Wires | Multiple | 5,000 | 5,000.0 |
| 10.0 | Ply Board | 1 | 5,000 | 5,000.0 |
| 11.0 | Pipes | 1 | 3,000 | 3,000.0 |
| 12.0 | Bucket | 1 | 2,500 | 2,500.0 |
| 13.0 | Bowl | 1 | 2,000 | 2,000.0 |
| 14.0 | LED | 1 | 200 | 200.0 |
| 15.0 | Breadboard | 1 | 5,000 | 5,000.0 |

| | | | | |
|---|---|---|---|---|
| 16.0 | Veroboard | 2 | 5,000 | 10,000.0 |
| 17.0 | Soldering Iron | 1 | 5,000 | 5,000.0 |
| 18.0 | Soldering Lead | 3 | 5,500 | 16,500.0 |
| 19.0 | Box Unit | 1 | 5,000 | 5,000.0 |
| 20.0 | Buzzer | 1 | 1,500 | 1,500.0 |
| 21.0 | Switch | 1 | 500 | 500.0 |
| 22.0 | LM7805 | 1 | 3,500 | 3,500.0 |
| 23.0 | Logistics | | 10,000 | 10,000.0 |
| 24.0 | Miscellaneous | | 20,000 | 20,000.0 |
| | TOTAL | | | 138,300.0 |
| | CONTINGENCY (5%) | | | 6,915.0 |
| | GRAND TOTAL | | | 145,215.0 |

# CHAPTER 5

# CONCLUSION AND RECOMMENDATIONS

## 5.1 CONCLUSION

This project shows how to successfully design, implement, and test an inexpensive, Internet of Things-based smart agriculture automation and monitoring system. Addressing the major gaps in the literature review, namely, the absence of a system that is affordable, user-friendly, and remotely accessible from any device, was the main goal.

We successfully utilized low-cost hardware in the design of the system, hardware such as the ESP32-WROOM microcontroller, common sensors (DHT11, DS18B20, resistive soil moisture), and a basic relay. The development of a universal web application that is accessible from anywhere in the world was the primary innovation here. This front-end is built with standard HTML, CSS, and JavaScript and connects to a free Google Sheets backend. This design successfully gets around the two main limitations of existing research:

(i). Our system doesn't require a specific app or platform for anyone to access it.

(ii). It is an effective and cheap system and doesn't require high capital for its implementation.

The firmware of the system turned out to be reliable. The irrigation pump was successfully managed by the automation logic with a hysteresis loop, and the 20-second safety timeout successfully avoided possible overwatering. Thus, the project's goal of developing an accessible, cost-effective, and useful automation and monitoring system that is ideal for subsistence farmers has been accomplished.

## 5.2 Recommendations for Future Work

The following enhancements are suggested for a subsequent iteration of this project in light of these findings:

(i). **Upgrade the Soil Sensor:** A capacitive soil moisture sensor ought to be used in place of the resistive one. Because capacitive sensors measure moisture without direct electrical contact and lack exposed electrodes, they are impervious to corrosion and greatly increase system reliability.

(ii). **Add Offline Data Logging:** The ESP32 needs an SD card module in order to overcome the network dependency. This would enable the gadget to locally store all sensor data. To make sure no data is ever lost, the ESP32 could upload this buffered data to the Google Sheet once the Wi-Fi connection is restored.

(iii). **Install a Solar Power System:** The 7.4V battery pack needs to be connected to a solar panel and a charge controller based on the TP4056 in order to produce a field unit that is completely self-sustaining. The farmer wouldn't have to manually recharge the battery as a result.

(iv). **Boost Power Efficiency:** A switching (buck) converter should be used in place of the ineffective LM7805 regulator. The battery life of the system would be significantly extended by a buck converter, which is far more effective at stepping down the voltage.

(v). **Improve the Web App:** Remote control should be added to the web application. To enable the farmer to manually turn the pump on or off from their phone, overriding the automation logic, when necessary, this would entail sending a command from the web app back to the ESP32 (for example, via the Google Sheet).

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## APPENDIX

```
// ESP32 Plant Monitor - Relay version

#include <WiFi.h>

#include <HTTPClient.h>

#include <DHT.h>

#include <OneWire.h>

#include <DallasTemperature.h>

#include <LiquidCrystal_I2C.h>

// ====== CONFIG ======

const char* WIFI_SSID = "MjStartlink1";

const char* WIFI_PASS = "maega299";

const char* scriptUrl =
"https://script.google.com/macros/s/AKfycbxMVVuhUT3m9h3LFxj4tJloruoqIPcapuf5XSamDt
NK1vnTorkiXxhORyAF_AsDZ9K2/exec"; // replace

// Pins

#define DHTPIN 4

#define DHTTYPE DHT11

#define ONE_WIRE_BUS 16

#define SOIL_PIN 36     // ADC1_CH0

#define RELAY_PIN 26    // IN pin on relay module

// Objects

DHT dht(DHTPIN, DHTTYPE);

OneWire oneWire(ONE_WIRE_BUS);
```

```
DallasTemperature ds18b20(&oneWire);
LiquidCrystal_I2C lcd(0x27, 16, 2);
// Timing
unsigned long lastSend = 0;
const unsigned long SEND_INTERVAL = 1UL * 60UL * 1000UL; // 5 minutes
unsigned long pumpActivatedAt = 0;
// Pump thresholds and safety
const int DRY_THRESHOLD = 2200;   // adjust after calibration (soilRaw > dry -> start pump)
const int WET_THRESHOLD = 1600;   // hysteresis (soilRaw < wet -> stop pump)
const unsigned long MAX_PUMP_RUN_MS = 20UL * 1000UL; // max run per activation (20 seconds)
// Relay active state
const bool RELAY_ACTIVE_LOW = true;
void setup() {
  Serial.begin(115200);
  Serial.println();
  Serial.println("===== ESP32 Automatic Plant Watering System =====");
  dht.begin();
  ds18b20.begin();
  lcd.init();
  lcd.backlight();
  pinMode(RELAY_PIN, OUTPUT);
```

```
  if (RELAY_ACTIVE_LOW) digitalWrite(RELAY_PIN, HIGH); else
digitalWrite(RELAY_PIN, LOW);
  // WiFi connect
  lcd.setCursor(0,0);
  lcd.print("WiFi Connecting");
  WiFi.begin(WIFI_SSID, WIFI_PASS);
  Serial.print("Connecting to WiFi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(400);
    Serial.print(".");
  }
  Serial.println();
  Serial.print("Connected! IP address: ");
  Serial.println(WiFi.localIP());
  lcd.clear();
  lcd.print("WiFi Connected");
  delay(900);
  lcd.clear();
}
void loop() {
  // Read sensors
  float airHumidity = dht.readHumidity();
  float airTemp = dht.readTemperature();
```

```
ds18b20.requestTemperatures();
float soilTemp = ds18b20.getTempCByIndex(0);
int soilRaw = analogRead(SOIL_PIN); // 0-4095
// 🔹 Added: detailed serial printout
Serial.println("-----------------------------------------------");
Serial.println("Sensor Readings:");
Serial.printf("Air Temp: %.1f °C\n", airTemp);
Serial.printf("Air Humidity: %.1f %%\n", airHumidity);
Serial.printf("Soil Temp: %.1f °C\n", soilTemp);
Serial.printf("Soil Moisture (raw): %d\n", soilRaw);
Serial.printf("Pump State: %s\n", isPumpOn() ? "ON" : "OFF");
Serial.println("-----------------------------------------------\n");
// Pump control with hysteresis & max run timeout
bool pumpIsOn = isPumpOn();
if (!pumpIsOn && soilRaw > DRY_THRESHOLD) {
  setPump(true);
  pumpActivatedAt = millis();
  Serial.println("Soil is dry -> Pump turned ON");
} else if (pumpIsOn && soilRaw < WET_THRESHOLD) {
  setPump(false);
  pumpActivatedAt = 0;
  Serial.println("Soil is wet -> Pump turned OFF");
```

```
  }
  // Safety timeout
  if (pumpIsOn && pumpActivatedAt > 0 && millis() - pumpActivatedAt >
MAX_PUMP_RUN_MS) {
    Serial.println("⚠️ Pump max runtime reached. Turning off.");
    setPump(false);
    pumpActivatedAt = 0;
  }
  // LCD update
  lcd.setCursor(0,0);
  lcd.print("A:");
  if (!isnan(airTemp)) lcd.print(airTemp,1); else lcd.print("--");
  lcd.print("C ");
  if (!isnan(airHumidity)) lcd.print(airHumidity,0); else lcd.print("--");
  lcd.print("%   ");
  lcd.setCursor(0,1);
  lcd.print("S:");
  if (!isnan(soilTemp)) lcd.print(soilTemp,1); else lcd.print("--");
  lcd.print("C ");
  lcd.print("M:");
  lcd.print(soilRaw);
  lcd.print(" ");
  lcd.print(isPumpOn() ? "PON" : "OFF");
```

```
  // 5-min send
  if (millis() - lastSend >= SEND_INTERVAL) {
    Serial.println("Sending data to Google Sheet...");
    sendToSheet(airTemp, airHumidity, soilTemp, soilRaw, isPumpOn());
    lastSend = millis();
  }
  delay(2000);
}
bool isPumpOn() {
  int out = digitalRead(RELAY_PIN);
  if (RELAY_ACTIVE_LOW) return (out == LOW);
  else return (out == HIGH);
}
void setPump(bool on) {
  if (RELAY_ACTIVE_LOW) {
    digitalWrite(RELAY_PIN, on ? LOW : HIGH);
  } else {
    digitalWrite(RELAY_PIN, on ? HIGH : LOW);
  }
  Serial.printf("Relay state setPump(%s)\n", on ? "ON" : "OFF");
}
void sendToSheet(float airT, float airH, float soilT, int soilRaw, bool pump) {
  if (WiFi.status() != WL_CONNECTED) {
```

```
  Serial.println("WiFi not connected -- skipping send");
  return;
}
HTTPClient http;
http.begin(scriptUrl);
http.addHeader("Content-Type", "application/json");
String payload = "{";
payload += "\"timestamp\":\""  + "\",";
payload += "\"dht_temp\":" + String(airT) + ",";
payload += "\"dht_humidity\":" + String(airH) + ",";
payload += "\"ds18b20_temp\":" + String(soilT) + ",";
payload += "\"soil_raw\":" + String(soilRaw) + ",";
payload += "\"pump_state\":" + String(pump ? 1 : 0);
payload += "}";
int code = http.POST(payload);
if (code > 0) {
  String resp = http.getString();
  Serial.printf("✅ POST code %d, resp: %s\n", code, resp.c_str());
} else {
  Serial.printf("❌ POST failed, err: %s\n", http.errorToString(code).c_str());
}
http.end();
```

```
}
```